Hardware-Aware Speculative KV-Cache Paging and Epistemic Attention Eviction for High-Throughput Multi-Turn Agentic LLM Serving

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***Abstract*—The rapid proliferation of autonomous multi-turn agentic workflows has exacerbated the memory capacity bottleneck in Large Language Model (LLM) inference accelerators. In modern agentic sequences spanning 32k to 128k+ tokens ($L = 64, H_{\text{kv}} = 8, d_k = 128$, `float16`), Key-Value (KV) cache memory footprints expand to 17.18 GB (16.00 GiB) per active 64k-token agent stream (8.59 GB / 8.00 GiB at 32k tokens). Even under multi-chip tensor-parallel weight and KV-cache sharding or 32k–64k context distributions, this memory demand precipitates an acute capacity cliff on accelerators such as the Google Cloud TPU v6e Trillium (32 GB HBM per chip), exhausting non-weight HBM capacity after only 3 to 8 concurrent sequences. Existing virtualized inference engines (e.g., PagedAttention / vLLM and SGLang / RadixAttention) mitigate Out-Of-Memory exceptions via reactive demand paging to Host DRAM over PCIe/CXL. However, servicing a synchronous full-layer demand page fault (64×64 KiB $= 4$ MiB across $L = 64$ layers, or $1.87\,\mu$s of raw wire serialization per single-layer 64 KiB block at 35 GB/s effective PCIe Gen5 bandwidth plus descriptor ring wakeup and dirty writeback) incurs severe pipeline stalls (0.12 ms to 0.50 ms per faulted step against 0.16 ms ideal compute), causing matrix execution units to idle for 40% to 70% of execution time. Furthermore, classical page eviction policies (LRU/LFU) are attention-blind: they inadvertently evict initial prompt attention sinks and active tool schemas while preserving ephemeral chain-of-thought scratchpad tokens.**

To overcome these constraints, we present Speculative Epistemic KV-Cache Paging (SE-Paging) governed by the Epistemic Least-Likely-Attended (ELLA) eviction protocol. SE-Paging couples the agent's semantic execution graph and turn-smoothed epistemic branch entropy $\mathcal{H}_i(t)$ to physical memory management: (1) an Epistemic Working Set Estimator (EWSE) projects future attention mass $\mathcal{A}(P_k, t)$, permanently pinning attention sinks and retaining active/idle tool schemas and DAG-boosted history while decaying idle tool schemas below DAG-predicted history within ≈ 8 steps (≈ 1 turn) and aggressively evicting resolved scratchpad pages; (2) an Asynchronous DMA Speculative Prefetcher (ADSP) with sealed-page write-through host mirroring overlaps Host-to-Device page transfers with active matrix computation across a $\tau_{\text{lookahead}} = 2$-step horizon; and (3) ELLA establishes bounded competitive ratios relative to offline Belady MIN under both non-cascading trace-regular greedy eviction and randomized Predictive Marker phases. We evaluate SE-Paging in a downscaled $1 : 32$ trace-driven CPU-based physical tensor-buffer memory-hierarchy emulator (modeling a 2-layer 128 KiB slice per $B = 16$-token page, $N_{\text{dev}} = 96$ HBM pages [12.00 MiB], $N_{\text{host}} = 2{,}500$ DRAM pages [312.50 MiB], and 976 allocated physical pages [16 shared prefix + 48 private pages/stream, or 15,616 total tokens] across $K = 20$ streams) parameterized to scaled TPU v6e Trillium and PCIe Gen5 timing, complemented by a 6-turn live trace replay on Google Cloud Vertex AI (`gemini-2.5-flash`) and parameterized multi-turn SWE-bench and Gorilla API emulation traces. Across 5,940 logical page-lookup cycles (11,880 two-stage memory-subsystem operations per protocol) spanning $K \in \{4, 8, 12, 16, 20\}$ concurrent agent streams under $10.17\times$ physical ($13.33\times$ unshared logical) working-set overflow, SE-Paging elevates cache hit rate at $K = 20$ from 0.20% (LRU) and 1.21% (LFU) to 28.23% ($28.31 \pm 0.38\%$ 5-seed mean), retains 98.64% empirical Query-Key (QK) RoPE dot-product attention mass, eliminates 46.8% of memory pipeline stall overhead, and increases simulated device throughput by 49.3% (+53.4% in host wall-clock page-lookup throughput) over LRU.

***Index Terms*—KV-Cache Memory Management, Hardware Accelerators, TPU v6e Trillium, Speculative DMA Prefetching, Multi-Agent LLM Serving, PagedAttention, Virtual Memory.**

I. Introduction

Enterprise deployment of autonomous agent swarms built on foundation models (e.g., Gemini 2.5 Pro/Flash, Gemma 2, and Claude) represents a major workload shift in cloud computing. In frameworks such as the Google Agent Development Kit (ADK) and Vertex AI Agent Builder, agents engage in prolonged, multi-turn reasoning workflows involving hierarchical sub-goal decomposition, external tool invocations via OpenAPI schemas [18], and structured artifact generation [1], [2].

A. The KV-Cache HBM Capacity Cliff

Unlike traditional stateless NLP classification or single-turn dialogue, agentic sequences accumulate extensive conversational histories, tool schemas, and chain-of-thought scratchpads, routinely expanding context lengths from 32k to 128k+ tokens [3], [4].

During auto-regressive transformer decoding, Key and Value activation vectors for every past token must remain memory-resident to avoid redundant quadratic recomputation. The physical memory required scales linearly with context length S, transformer layers L, KV heads H_{kv}, and head dimension d_k:

$$\text{Memory}_{\text{KV}} = 2 \times L \times H_{\text{kv}} \times d_k \times S \times \text{bytes_per_elem} \quad (1)$$

For a representative enterprise model ($L = 64$, $H_{\text{kv}} = 8$, $d_k = 128$, `float16` with bytes_per_elem $= 2$), a sequence

of $S = 64\text{k} = 65{,}536$ tokens requires $17{,}179{,}869{,}184$ bytes $\approx$ $\mathbf{17.18}$ GB decimal ($\mathbf{16.00}$ GiB binary) of dedicated High-Bandwidth Memory (HBM) (or $8{,}589{,}934{,}592$ bytes $\approx$ $\mathbf{8.59}$ GB decimal / $\mathbf{8.00}$ GiB binary for $S = 32\text{k} = 32{,}768$ tokens).

On modern cloud accelerators such as the Google Cloud TPU v6e Trillium (configured with 32 GB HBM in 256-chip 2D torus pods, following Google's custom TPU architectures [5]) or GPU nodes [6], this memory demand triggers an immediate *capacity cliff*: even under multi-chip tensor-parallel model-weight and KV-cache sharding (where 64-layer weights are partitioned across chips, leaving 12–24 GiB of non-weight HBM per chip) or 32k–64k context distributions with shared prefixes, a chip exhausts its remaining HBM pool after serving only 3 to 8 concurrent agent sequences, capping cluster concurrency.

B. The Demand Paging Pipeline Stall Dilemma

To prevent fatal Out-Of-Memory (OOM) aborts, modern serving runtimes (e.g., vLLM / PagedAttention [3] and SGLang [1]) partition the KV-cache into discrete non-contiguous physical pages ($B = 16$ tokens per block) and implement reactive *demand swapping* to Host DRAM over PCIe Gen5 or CXL, while multi-turn offloading systems such as CachedAttention [2] perform coarse-grained inter-turn session pre-loading across conversation boundaries. Because attending densely to all historical blocks on every token step under $> 10\times$ HBM overcommit would force continuous PCIe thrashing over completed reasoning steps, production long-context agentic runtimes increasingly pair virtualized paging with *selective segment-routed* or *router-guided working-set paging*, where each decoding step attends over persistent attention sinks ($\mathcal{P}_{\text{anchor}}$), the executing stream's active local window ($\mathcal{P}_{\text{active},i^*(t)}$), and the orchestrator-routed active working-set segments ($\mathcal{W}_{\text{active_seg}} \subseteq \mathcal{P}_{\text{tool}} \cup \mathcal{P}_{\text{hist}} \cup \mathcal{P}_{\text{scratch}}$).

However, even under selective working-set routing, intra-turn demand swapping remains fundamentally reactive: when the accelerator's matrix execution units encounter an unmapped tool-schema or history page during token generation, the decoding pipeline stalls. Over a PCIe Gen5 $\times 16$ interconnect (64 GB/s theoretical unidirectional bandwidth, $\approx$ 35 GB/s effective under multi-stream contention), transferring a single-layer 64 KiB page block ($B = 16, H_{\text{kv}} = 8, d_k = 128$) requires $\approx 1.87\,\mu$s of raw wire serialization ($3.74\,\mu$s per fused 2-layer 128 KiB slice, or $119.7\,\mu$s across all $L = 64$ transformer layers totaling 4 MiB per 16-token block); combined with PCIe DMA descriptor ring wakeup and synchronous dirty-page writeback, a full-model page fault stalls token generation for 0.12 **ms to** 0.50 **ms** (0.288–0.384 ms in our calibrated parameterization vs. 0.160 ms ideal 64-layer compute) [6], [7]. Under high sequence concurrency, these synchronous demand stalls accumulate, causing matrix execution units to spend **40% to 70% of wall-clock time stalled waiting on memory**.

C. The Attention-Blind Eviction Failure

Furthermore, classical operating system page eviction policies (Least Recently Used [LRU], Least Frequently Used [LFU], FIFO) and existing token-pruning or single-step rehearsal heuristics are mismatched with multi-turn agentic transformer physics:

1) **Destruction of Attention Sinks and Tool Schemas:** Transformers concentrate massive attention mass on the initial 1 to 4 sink blocks [4] and active OpenAPI tool definitions [18]. Under LRU, as an agent generates thousands of intermediate reasoning tokens, initial prompt tokens and tool schemas are aged out to Host RAM. While static window methods like StreamingLLM [4] drop middle tokens permanently, Heavy-Hitter Oracle (H2O) [8] tracks cumulative attention scores, Quest [9] estimates page criticality from query-key bounding boxes within the current layer without multi-step DMA prefetching, and InfiniGen [7] performs single-step layer-adjacent rehearsal prefetching (using skewed query-key projections in layer $l-1$ to prefetch layer l), none of these approaches satisfy multi-step PCIe lookahead horizons ($\tau_{\text{lookahead}} = 2$ steps ahead) during abrupt multi-turn agentic tool-call DAG transitions where an entire dormant tool schema or prior history segment suddenly regains high query-key attention alignment. Similarly, unlike purely static agentic workflow prefetchers (KVFlow [10]) or prefix-tree LRU schedulers (SGLang / RadixAttention [1]) that ignore live generation uncertainty, SE-Paging couples workflow DAG lookahead with turn-smoothed epistemic Shannon entropy $\mathcal{H}_i(t)$.
2) **Scratchpad Memory Pollution:** Intermediate chain-of-thought scratchpad tokens are predominantly transient: once an agent concludes a reasoning sub-step, its scratchpad tokens carry rapidly decaying query-key weight unless backtracking occurs. Classical LRU retains newly written scratchpad pages in fast HBM, displacing reusable history and tool schemas.

D. Contributions

In this paper, we propose **Hardware-Aware Speculative KV-Cache Paging (SE-Paging)** governed by the **Epistemic Least-Likely-Attended (ELLA)** eviction protocol for router-guided two-tier working-set paging. Our key contributions are:

1) **Epistemic Working Set Estimator (EWSE):** A formal priority formulation that projects future page attention mass $\mathcal{A}(P_k, t)$ from agent semantic segment tags, stream-indexed tool-call and history DAG lookahead $\hat{d}_{\text{DAG},i}(P_k, t)^{-1}$, and conditionally re-normalized action-delimiter epistemic Shannon entropy $\mathcal{H}_i(t)$.
2) **Asynchronous DMA Speculative Prefetcher (ADSP):** A non-blocking prefetcher restricted to lookahead-predicted non-resident candidates with sealed-page write-through host mirroring that dispatches Host-to-Device transfers at step $t = n - \tau_{\text{lookahead}}$, overlapping

PCIe/CXL transfer latency with matrix compute while eliminating synchronous dirty-page writebacks.

3) **Theoretical Performance Guarantees:**
   - **Theorem 1 (Queued Zero-Stall Pipeline Overlap):** Establishes the step-indexed multi-page lookahead condition $\tau_{\text{lookahead}} \geq \left\lceil \frac{M_{\text{pre}}(n) T_{\text{DMA}}(P_k)}{(1-\rho_{\text{bus}}) T_{\text{compute}}} \right\rceil$ under an unbacklogged or steady-state stable DMA queue under which device stall cycles are identically zero.
   - **Theorem 2 (Attention Mass Preservation):** Proves that ELLA scratchpad eviction beyond the pinned active window ($\delta = 64$ tokens) of the executing stream analytically bounds attention mass loss by $\epsilon_{\text{attn}} \leq C\delta^{1-\gamma}\left(\frac{1}{\delta} + \frac{1}{\gamma-1}\right) \leq \frac{C_{\text{scratch}}}{(\gamma-1)\delta^{\gamma-1}}$ (≤ 0.0261 at $\delta = 64$ and ≤ 0.0105 at $\delta = 256$ from the intermediate integral bound), complemented by non-circular empirical Query-Key (QK) RoPE softmax attention evaluation showing 98.64% retained attention mass (96.80% top-k QK page recall).
   - **Theorem 3 (Bounded Competitive Ratio):** Uses the competitive caching with predictions framework [11] on acyclic agentic execution graphs to establish expected competitive ratios of $\mathbb{E}[\mathcal{C}_{\text{ELLA}}^{\text{greedy}}] \leq 1+\mathcal{O}(\eta_{\text{tot}}/\text{OPT}+\mathcal{H}_{\max})$ under non-cascading trace-regular greedy eviction and $\mathbb{E}[\mathcal{C}_{\text{ELLA}}^{\text{marker}}] \leq 2 + \mathcal{O}(\min(\sqrt{\mathbb{E}[\eta_{\ell_1}]/\text{OPT}}, \log N_{\text{dev}}))$ when paired with randomized Predictive Marker phases relative to offline Belady MIN [12].

4) **Downscaled Physical Tensor-Buffer Emulator, Multi-Turn Benchmark Generalization & Live Vertex AI Trace Replay:** We evaluate SE-Paging in a downscaled $1 : 32$ CPU-based physical tensor-buffer emulator (modeling a 2-layer 128 KiB slice per page, $N_{\text{dev}} = 96$ HBM pages [12.00 MiB], $N_{\text{host}} = 2{,}500$ DRAM pages [312.50 MiB], and 976 allocated physical pages [16 shared prefix + 48 private pages/stream, or 15,616 total tokens] across $K = 20$ streams) across 5,940 logical page-lookup cycles (11,880 two-stage memory-subsystem operations per protocol), comparing against 11 baseline and ablation configurations (`DEMAND_LRU`, `DEMAND_LFU`, `SGLang/RadixAttention` [1], `StreamingLLM` [4], `H2O` [8], `Quest` [9], `CachedAttention` [2], `InfiniGen` [7], `DEMAND_ELLA` with and without write-through mirroring, and `STATIC_DAG_ELLA` [10]), plus a $\pm 50\%$ hyperparameter sensitivity study, a heterogeneous 20-prompt multi-tenant workload, parameterized multi-turn SWE-bench [17] and Gorilla OpenAPI [18] (stateful BFCL-style multi-turn tool-switching) emulation traces, and a 6-turn live Google Cloud Vertex AI (`gemini-2.5-flash`) trace replay.

II. Related Work

- **Virtualized KV-Cache and Offloading Systems:** PagedAttention / vLLM [3] introduced OS-inspired block-level KV-cache virtualization ($B = 16$ tokens) with reactive demand swapping, while SGLang [1] introduced RadixAttention for shared prefix-tree reuse with LRU leaf eviction. FlexGen [6] formulated static linear-programming offload schedules across GPU, CPU DRAM, and disk for offline batch throughput, and CachedAttention [2] overlapped inter-turn session restoration across multi-turn conversation boundaries. However, none of these systems prevent intra-turn page-fault stalls when an active agent dynamically switches between tool schemas and history blocks across sub-steps.
- **Token Pruning and Intra-Layer Sparse Attention:** StreamingLLM [4] retains initial attention sinks plus a static sliding window while permanently discarding middle tokens. H2O [8] evicts tokens with low cumulative attention scores, Quest [9] selects top-k pages per layer using query-key bounding boxes without asynchronous Host-to-Device DMA lookahead, and InfiniGen [7] prefetches KV entries from layer $l-1$ to layer l via skewed projections. While effective for single-step layer overlap, single-layer lookahead ($\tau = 1$ layer) is insufficient to hide multi-page PCIe Gen5 transfer latency across multi-turn agentic state transitions.
- **Agent-Aware, Dynamic Workflow-Prediction, and Adaptive-TTL KV-Cache Management:** Recent agentic serving frameworks such as KVFlow [10] and SGLang [1] exploit static agentic workflow graphs or radix trees to pre-warm or pin agent prefixes during structured tool-calling workloads [17], [18]. Emerging dynamic workflow-caching and offloading architectures extend this paradigm along three axes: (i) prediction-based multi-agent workflow schedulers (e.g., PBKV) that estimate inter-agent invocation probabilities across DAG nodes to pre-stage downstream agent contexts; (ii) cost-aware adaptive Time-To-Live (TTL) schedulers (e.g., Continuum) that estimate external tool-execution pause durations between turns to set job-level HBM retention TTLs; and (iii) two-tier segment offloading engines (e.g., LMCache and CachedAttention [2]) alongside region-aware exponential prompt decay and recallable host-memory page eviction (evaluated in our fixed-TTL, HBM-utilization-throttled adaptive-TTL, and observation-window ablations in Section VI-C4). However, inter-agent invocation prediction and inter-turn pause TTLs operate at coarse macro-turn or cross-agent granularity: within a single agent's multi-step reasoning turn, when the model encounters high epistemic branching uncertainty during tool selection or error recovery, static DAG weights and turn-level TTL pins either retain resolved chain-of-thought scratchpad blocks unnecessarily, exhaust HBM when large tool catalogs (e.g., 32 tool-schema pages) are simultaneously pinned, or fail to trade off recent versus DAG-predicted history segments dynamically. SE-Paging bridges this gap by operating at intra-turn step granularity: it couples fine-grained semantic page classes ($\mathcal{P}_{\text{anchor}}, \mathcal{P}_{\text{tool}}, \mathcal{P}_{\text{active}}, \mathcal{P}_{\text{hist}}, \mathcal{P}_{\text{scratch}}$), asymptotic retention floors ($C_{\text{floor}}, C_{\text{hist}}$), continuous exponen-

tial idle-tool decay (λ_{tool}), and stream-indexed forward DAG distance $\hat{d}_{\text{DAG},i}(P_k,t)^{-1}$ with live action-delimiter epistemic Shannon entropy $\mathcal{H}_i(t)$ over grammar-valid schema tokens $\mathcal{V}_{\text{act}}$ and sealed-page write-through DMA mirroring.

III. System Architecture and Memory Hierarchy

A. Heterogeneous Memory Hierarchy Model

We model an AI inference accelerator node (e.g., Google Cloud TPU v6e Trillium programmed via JAX/Pallas kernels [13]) comprising two memory tiers:

- **Tier 0 (Device HBM):** High-bandwidth, capacity-constrained memory directly attached to Matrix Execution Units (MXUs) ($\text{BW}_{\text{HBM}} \geq 1.6$ TB/s, capacity $C_{\text{HBM}} \in \{32, 64\}$ GB).
- **Tier 1 (Host DRAM / CXL):** High-capacity system memory connected via PCIe Gen5 / CXL ($\text{BW}_{\text{bus}} \approx$ 32–64 GB/s, capacity $C_{\text{Host}} \geq 512$ GB).

KV-cache is managed in fixed-size physical page blocks P_k containing $B = 16$ tokens. For a single transformer layer ($H_{\text{kv}} = 8, d_k = 128$, `float16`), each page block occupies:

$$\begin{aligned}\text{Size}_{\text{1-layer}}(P_k) &= B \times 2 \times H_{\text{kv}} \times d_k \times 2 \text{ bytes} \\ &= 65{,}536 \text{ bytes } (64 \text{ KiB})\end{aligned} \tag{2}$$

In our multi-layer block allocator (Section VI-A), pairs of adjacent transformer layers are co-allocated into fused 128 KiB (131,072 byte) physical transfer units to amortize PCIe DMA descriptor ring overhead while permitting layer-pipelined DMA interleaving and sub-stack preemption across concurrent streams without blocking on a monolithic 4 MiB transfer. In a full 64-layer deployment, the 32 fused 2-layer slices of logical block P_k share a single logical page-table descriptor and priority score $\mathcal{A}(P_k,t)$ equipped with a 32-bit layer-residency bitmap (`slice_mask`); while uniform full-stack 32-slice block residency trades fine-grained layer-heterogeneous cache budgeting for $\mathcal{O}(1)$ page-table metadata overhead, if sub-stack preemption interrupts a 64-layer transfer after slice $s < 32$, layers $1 \ldots 2s$ execute while the remaining slices resume at the head of the DMA queue prior to layer $2s+1$, and partially resident blocks ($0 <$ popcount(`slice_mask`) < 32) remain pinned in $\mathcal{P}_{\text{pinned_prefetch}}$ until all 32 slices complete so partially populated blocks are never prematurely evicted. In our scaled $1:32$ slice emulator (modeling a representative 2-layer 128 KiB slice out of $L = 64$ layers), per-slice raw DMA wire transfer latency $T_{\text{DMA}}(P_k) = 3.74\,\mu$s ($4.56\,\mu$s effective slice transfer latency under $\rho_{\text{bus}} = 0.18$ background bus contention), per-slice ideal matrix compute time $T^{\text{ideal}}_{\text{compute}} = 5.00\,\mu$s, and scaled descriptor wakeup / writeback penalties ($T_{\text{wakeup}} = 4.44\,\mu$s, representing the $1:32$ emulator's amortized per-slice share of $142.08\,\mu$s full-stack host page-fault handling and DMA descriptor ring initiation for a non-preempted 32-slice transfer—where sub-stack preemption after slice $s < 32$ would incur an additional hardware ring wakeup upon resumption—and $T_{\text{wb}} = 3.00\,\mu$s, yielding $4.44 + 4.56 = 9.00\,\mu$s per clean slice fault or $32 \times (4.44 + 4.56) = 142.08 + 145.92 = 288.0\,\mu$s across 64 layers, and $4.44 + 4.56 + 3.00 = 12.00\,\mu$s per dirty slice fault or $384.0\,\mu$s across 64 layers) scale uniformly by $1:32$ relative to a full 64-layer step ($T^{\text{64L}}_{\text{DMA}} \approx 119.7\,\mu$s raw$/145.92\,\mu$s effective, $T^{\text{64L}}_{\text{compute}} = 160.0\,\mu$s), preserving exact dimensionless overlap ratios, page-replacement hit rates, and working-set overflow ratios.

B. Categorical Page Classification and Boundary Resolution

As tokens are ingested or generated by the orchestrator (e.g., Google ADK or SGLang [1]), lightweight structural delimiters (`<|system|>`, `<|tool_schema|>`, `<|history|>`, `<|thought|>`) tag each token span in $\mathcal{O}(1)$ time with a persistent underlying structural class ($\mathcal{P}_{\text{anchor}}, \mathcal{P}_{\text{tool}}, \mathcal{P}_{\text{hist}}, \mathcal{P}_{\text{scratch}}$), while dynamically maintaining an active sliding window ($\mathcal{P}_{\text{active}}$):

1) **System Anchor ($\mathcal{P}_{\text{anchor}}$):** System prompts and invariant instructions acting as persistent attention sinks [4].
2) **Tool Schema ($\mathcal{P}_{\text{tool}}$):** OpenAPI function declarations and JSON schema contracts. Shared tool schemas ($\mathcal{P}^{\text{shared}}_{\text{tool}}$) reside in the fixed-offset 16-page swarm prefix ($4\ \mathcal{P}_{\text{anchor}} + 12\ \mathcal{P}^{\text{shared}}_{\text{tool}}$, positions $0 \ldots 255$) across all K streams so pre-computed RoPE key embeddings remain position-invariant across streams (whereas private dynamically injected tool schemas $\mathcal{P}^{\text{priv}}_{\text{tool}}$ occupy stream-local page descriptors).
3) **Active Generation Window ($\mathcal{P}_{\text{active}}$):** A temporary sliding-window protection applied to the 4 most recent $B = 16$-token blocks ($W_{\text{local}} = 64$ tokens at block-aligned boundaries, comprising the 1 unsealed active tail block in $\mathcal{P}_{\text{active_unsealed}}$ at $\Delta t_{\text{win}} = 0$ plus the 3 most recently sealed blocks at $\Delta t_{\text{win}} \in \{1,2,3\}$) of the currently executing stream $S_{i^*(t)}$.
4) **Conversation History ($\mathcal{P}_{\text{hist}}$):** Verified multi-turn user queries and committed tool outputs.
5) **Intermediate Scratchpad ($\mathcal{P}_{\text{scratch}}$):** Ephemeral chain-of-thought reasoning tokens from completed turns.

Dominant-Class Boundary and Non-Demoting Sliding-Window Rule: The page allocator supports two boundary modes: (i) *contiguous sub-block packing with dominant-class promotion* (used in our Section VI benchmarks so every sealed page maintains full $B = 16$-token occupancy and zero internal fragmentation), where any $B = 16$-token block straddling two delimiter spans dynamically promotes its persistent class $c(P_k) \in \{\mathcal{P}_{\text{anchor}}, \mathcal{P}_{\text{tool}}, \mathcal{P}_{\text{hist}}, \mathcal{P}_{\text{scratch}}\}$ under the dominance hierarchy $\mathcal{P}_{\text{anchor}} \succ \mathcal{P}_{\text{tool}} \succ \mathcal{P}_{\text{hist}} \succ \mathcal{P}_{\text{scratch}}$ upon filling 16 tokens; and (ii) *strict delimiter sealing*, where structural delimiter transitions immediately seal the current block prior to reaching 16 tokens (exiting $\mathcal{P}_{\text{active_unsealed}}$ and triggering background write-through mirroring to Host DRAM) so ephemeral scratchpad tokens never share a physical frame with higher-tier segments. In both modes, all 4 blocks of the currently executing stream's active window $\mathcal{P}_{\text{active},i^*(t)}$ (the 1 unsealed tail block at $\Delta t_{\text{win}} = 0$ plus the 3 most recently sealed blocks at $\Delta t_{\text{win}} \in \{1,2,3\}$ in W_{local}) are structurally excluded from $\mathit{Resident}_{\text{evictable}}(\mathit{HBM})$ during step t (Invariant 1) and assigned active-window score $\mathcal{A}_{\text{active},i}(P_k,t) \triangleq$

$6 \times 10^8 \exp(-\kappa \Delta t_{\text{win}}) \mathbb{I}[P_k \in \mathcal{P}_{\text{active},i} \wedge i = i^*(t)]$ via $\tilde{\mathcal{A}}_i(P_k, t) = \max(\mathcal{A}_{\text{class},i}(P_k, t), \mathcal{A}_{\text{active},i}(P_k, t))$. Setting the active-window base weight to $6 \times 10^8 > 5 \times 10^8$ ensures that an anchor (10^9) inside W_{local} is never demoted, while sealed W_{local} blocks of the executing stream $S_{i^*(t)}$ strictly outrank peer streams' active tool schemas (5×10^8) and revert to $\mathcal{A}_{\text{class},i}(P_k, t)$ when $S_{i^*(t)}$ yields or advances past 4 blocks. This prevents $K \times 3 = 60$ dormant sealed window blocks across $K = 20$ streams from monopolizing the 96-page HBM arena; if a yielded stream S_i's sealed local-window scratchpad block is evicted while dormant, ADSP automatically includes any non-resident block of $\mathcal{P}_{\text{active},i}$ in S_i's lookahead prefetch batch at step $t = n - \tau_{\text{lookahead}}$ (or, if capacity-gated, services it as a synchronous pre-step demand fault accounted directly in $N_{\text{miss}}^{\text{clean}}$ at $9.00\,\mu$s/slice prior to local-window attention).

IV. The Speculative Epistemic Paging Protocol

A. Epistemic Attention Scoring Formulation

Definition 1 (Projected Attention Score)**.** *Let $t \in \mathbb{N}$ denote the global execution / page-lookup step index scheduled round-robin across the K active streams, and let $t_i = \lfloor t/K \rfloor$ denote the stream-local step index of stream S_i (where each macroscopic agentic reasoning turn spans 8–12 stream-local execution steps). For a KV-cache page P_k evaluated for agent stream S_i at step t, let $\Delta t_i(P_k, t) = t_i - t_{\textit{last_access},i}(P_k) \geq 0$ denote the elapsed stream-local step count since the semantic segment containing P_k was last generated, selected by the router, or accessed by stream S_i (where $t_{\textit{last_access},i}(P_k)$ records the stream-local step index of S_i's most recent reference; equivalently, when indexed by global step t, the per-step decay rates $\lambda, \lambda_{\textit{tool}}, \mu$ are scaled by $1/K$ so per-turn decay is invariant across concurrency levels K), and let $\hat{d}_{\textit{DAG},i}(P_k, t) \in \mathbb{Z}_{\geq 1} \cup \{\infty\}$ denote the shortest-path positive stream-local forward step distance along stream S_i's mode-selected workflow branch ($\arg\max_{v \in \mathcal{V}_{\textit{act}}} \tilde{p}_{i,m}(v)$, or minimum distance across the top-$B_{\textit{top}} = 2$ branches when $\mathcal{H}_i(t) > 0.50$) from its current node to the next future node referencing P_k's segment. Concretely, the orchestrator maintains a lightweight segment-adjacency lookahead table indexed by candidate action token $v \in \mathcal{V}_{\textit{act}}$ that maps each branch decision v to: (a) its immediate ($\hat{d}^{(v)}_{\textit{DAG},i}(P_k, t) = 1$) OpenAPI tool-schema pages $P_k \in \mathcal{P}_{\textit{tool}}$, (b) downstream ($\hat{d}^{(v)}_{\textit{DAG},i}(P_k, t) \in \{2, 3, \dots\}$) tool-transition schemas in the declared ADK workflow graph, and (c) prior-turn conversation history segments $P_k \in \mathcal{P}_{\textit{hist}}$ linked by data-dependency edges (e.g., the prior tool-output or user-specification segment consumed by action v, or the parent turn segment targeted by a verification/backtrack delimiter). Its per-stream class score $\mathcal{A}_{\textit{class},i}(P_k, t)$ under ELLA is defined as:*

$$\mathcal{A}_{\textit{class},i}(P_k, t) \triangleq \begin{cases} 10^9, \textit{ if } c(P_k) = \mathcal{P}_{\textit{anchor}} \\ 5 \times 10^8 \, \mathbb{I}_{\textit{tool},i}(P_k, t) + (1 - \mathbb{I}_{\textit{tool},i}(P_k, t)) \\ \quad \times \Big(W_{\textit{tool}} e^{-\lambda_{\textit{tool}} \Delta t_i(P_k, t)} + C_{\textit{floor}} + \eta_{\textit{DAG}} \hat{d}_{\textit{DAG},i}(P_k, t)^{-1} \Big), \\ \textit{if } c(P_k) = \mathcal{P}_{\textit{tool}} \\ \max\Big(C_{\textit{hist}}, \, W_{\textit{hist}} e^{-\lambda \Delta t_i(P_k, t)} (1 - \mathcal{H}_i(t) + \epsilon) \Big) \\ \quad + B_{\textit{DAG}} \mathbb{I}_{\textit{DAG},i}(P_k, t) + \eta_{\textit{DAG}} \hat{d}_{\textit{DAG},i}(P_k, t)^{-1}, \\ \textit{if } c(P_k) = \mathcal{P}_{\textit{hist}} \\ W_{\textit{scratch}} \exp(-\mu \Delta t_i(P_k, t)) \, \mathcal{H}_i(t) + \eta_{\textit{DAG}} \hat{d}_{\textit{DAG},i}(P_k, t)^{-1}, \\ \textit{if } c(P_k) = \mathcal{P}_{\textit{scratch}} \end{cases} \tag{3}$$

with effective per-stream score $\tilde{\mathcal{A}}_i(P_k, t) = \max(\mathcal{A}_{\textit{class},i}(P_k, t), \mathcal{A}_{\textit{active},i}(P_k, t))$ using $\mathcal{A}_{\textit{active},i}(P_k, t)$ from Section III-B, and global shared-page priority across all referencing streams $\mathcal{O}(P_k)$:

$$\mathcal{A}(P_k, t) \triangleq \max_{S_i \in \mathcal{O}(P_k)} \tilde{\mathcal{A}}_i(P_k, t) \tag{4}$$

where $\mathbb{I}_{\textit{tool},i}(P_k, t) = \mathbb{I}[c(P_k) = \mathcal{P}_{\textit{tool}} \wedge \hat{d}_{\textit{DAG},i}(P_k, t) \leq d_{\textit{look}}] \in \{0, 1\}$ and $\mathbb{I}_{\textit{DAG},i}(P_k, t) = \mathbb{I}[c(P_k) = \mathcal{P}_{\textit{hist}} \wedge \hat{d}_{\textit{DAG},i}(P_k, t) \leq d_{\textit{look}}] \in \{0, 1\}$ equal 1 when the orchestrator's workflow DAG or grammar-constrained router predicts a reference to tool schema or history page P_k within stream-local lookahead depth $d_{\textit{look}} = 2$ steps (and 0 otherwise); $W_{\textit{tool}} = 500$, $W_{\textit{hist}} = 600$, and $W_{\textit{scratch}} = 100$ are the initial class weights; $C_{\textit{floor}} = 200$ and $C_{\textit{hist}} = 120$ are the asymptotic tool-schema and verified-history retention floors; $B_{\textit{DAG}} = 350$ is the history lookahead boost; $\hat{d}_{\textit{DAG},i}(P_k, t)^{-1} \in [0, 1]$ is the inverse forward step distance (0 if $\hat{d}_{\textit{DAG},i}(P_k, t) = \infty$, scaled by $\eta_{\textit{DAG}} = 10$ across all non-anchor tiers so an idle tool schema at $\hat{d}_{\textit{DAG},i} = 3$ steps ahead outranks permanently dormant tools); and stream-local step decay hyperparameters are calibrated on held-out multi-turn traces to $\kappa = 0.05$ block^{-1} (over window block offset $\Delta t_{\textit{win}} \in \{0, 1, 2, 3\}$), $\lambda = 0.08$ step^{-1}, $\lambda_{\textit{tool}} = 0.08$ step^{-1}, $\mu = 0.45$ step^{-1}, and $\epsilon = 0.10$ ($\mu \gg \lambda = \lambda_{\textit{tool}} > \kappa > 0$). For shared pages ($|\mathcal{O}(P_k)| > 1$), let $S_{i^\dagger(P_k,t)} \in \arg\max_{S_i \in \mathcal{O}(P_k)} \tilde{\mathcal{A}}_i(P_k, t)$ denote the maximizing owner stream in Eq. (4); ties in $\mathcal{A}(P_k, t)$ (and any internal co-maximizer ties for $i^\dagger$) are broken deterministically by $\hat{d}_{\textit{DAG},i^\dagger}(P_k, t)^{-1}$ (evicting smaller $\hat{d}^{-1}_{\textit{DAG},i^\dagger}$ first during victim selection and admitting larger $\hat{d}^{-1}_{\textit{DAG},i^\dagger}$ first during prefetching), then maximizing owner stream index $i^\dagger$, oldest stream-specific LRU timestamp $t_{\textit{last_access},i^\dagger}(P_k)$, and finally lowest physical page index k.

Idle Tool-Schema Decay Horizon vs. DAG-Boosted History Preemption: Under Eq. (3), a newly used tool schema entering the idle state ($\mathbb{I}_{\text{tool},i}(P_k, t) = 0$ at $\Delta t_i = 0, \hat{d}_{\text{DAG},i} = \infty$) has score $W_{\text{tool}} + C_{\text{floor}} = 500 + 200 = 700$, protecting it against immediate eviction during the current turn. Once dormant for

$\Delta t_i > \ln(500/270)/0.08 \approx 7.70$ stream-local steps (≈ 1 turn at 8–12 stream-local steps/turn), its priority drops below 470 (reaching 300.9 by $\Delta t_i = 20$ stream-local steps), enabling a floored DAG-predicted history page ($\geq C_{\text{hist}} + B_{\text{DAG}} = 120 + 350 = 470$, or 480 at $\hat{d}_{\text{DAG},i} = 1$) to preempt dormant tool schemas when resuming an earlier conversation branch. Meanwhile, because $t_{\text{last_access},i}(P_k)$ and the segment-adjacency lookahead table for $\hat{d}_{\text{DAG},i}(P_k, t)$ are maintained at segment-to-page table granularity by the orchestrator (requiring at most $|\mathcal{V}_{\text{act}}| = 64$ table lookups at action-delimiter steps), descriptor state updates require $\mathcal{O}(1)$ time per page access (paired with an $\mathcal{O}(N_{\text{dev}})$ priority scan over the $N_{\text{dev}} = 96$ HBM descriptors upon victim selection), avoiding any $\mathcal{O}(S)$ token-level scan or quadratic attention matrix materialization inside fused Pallas FlashAttention-2 kernels [14].

Action-Delimiter Epistemic Entropy $\mathcal{H}_i(t)$ and Multi-Branch Lookahead Coupling: To prevent deterministic JSON punctuation and syntax tokens from driving entropy artificially toward zero, the orchestrator evaluates Shannon entropy exclusively when step t coincides with a structural branch or action delimiter $m(t) \in \mathcal{T}_{\text{branch}}$ (e.g., `<|thought|>`, `<|tool_call|>`, and tool-name selection tokens) over the grammar-valid candidate action-token subset $\mathcal{V}_{\text{act}}$ ($|\mathcal{V}_{\text{act}}| = 64$, or the prompt's fixed active schema set $\mathcal{V}_{\text{act}}^{\text{API}}$ intersected with returned Top-$K_{\log}$ logprobs on commercial APIs), holding $\mathcal{H}_i(t) = \mathcal{H}_i(t-1)$ constant across non-delimiter decoding steps (while physical block sealing occurs every $B = 16$ generated tokens or upon a delimiter transition). Raw token probabilities are conditionally re-normalized over $\mathcal{V}_{\text{act}}$ with Laplace smoothing floor $\epsilon_{\text{prob}} = 10^{-4}$ (applied when any candidate in $\mathcal{V}_{\text{act}}$ falls outside truncated API top-$K_{\log}$ logprobs, and reducing to $\epsilon_{\text{prob}} = 0$ when full logits over $\mathcal{V}_{\text{act}}$ are available) via $\tilde{p}_{i,m}(v) = \max(p_{i,m}(v), \epsilon_{\text{prob}}) / \sum_{u \in \mathcal{V}_{\text{act}}} \max(p_{i,m}(u), \epsilon_{\text{prob}})$ so that $\sum_{v \in \mathcal{V}_{\text{act}}} \tilde{p}_{i,m}(v) = 1$, and smoothed from neutral prior $\mathcal{H}_i(0) = 0.50$ via an exponential moving average with factor $\beta = 0.30$:

$$\mathcal{H}_i(t) = (1-\beta)\mathcal{H}_i(t-1) + \beta\left(-\frac{1}{\ln|\mathcal{V}_{\text{act}}|}\sum_{v \in \mathcal{V}_{\text{act}}} \tilde{p}_{i,m}(v) \ln \tilde{p}_{i,m}(v)\right) \in [0,1] \tag{5}$$

with $0 \ln 0 \triangleq 0$. While strict tier-bound separation (maximum unwindowed scratchpad score $W_{\text{scratch}} + \eta_{\text{DAG}} = 110 <$ asymptotic floors $C_{\text{hist}} = 120 < C_{\text{floor}} = 200$) guarantees that neither dormant tool schemas nor verified history ever decay below unwindowed scratchpad pages as $\Delta t_i \to \infty$, modulating $\mathcal{H}_i(t) \in [0,1]$ drives three synergistic mechanisms: (1) *intra-tier history crossover*, where high branch uncertainty ($\mathcal{H}_i(t) \to 1$) rapidly anneals unpredicted recent history pages $W_{\text{hist}} e^{-\lambda \Delta t_i}(1 - \mathcal{H}_i(t) + \epsilon)$ toward $C_{\text{hist}} = 120$, allowing older DAG-boosted history pages (≥ 470) to outrank recent turns during exploratory backtracking; (2) *cross-stream scratchpad arbitration*, where across concurrent streams $S_i \neq S_j$, stream S_i in high-uncertainty exploration ($\mathcal{H}_i(t) \to 1$) retains higher scratchpad priority than stream S_j whose branching has resolved ($\mathcal{H}_j(t) \to 0$ via geometric EMA decay $(1-\beta) = 0.7$ per delimiter, where scratchpad priority decays toward ≤ 10, i.e., $\eta_{\text{DAG}} \hat{d}_{\text{DAG},i}(P_k, t)^{-1}$); and (3) *entropy-gated runner-up branch protection and speculative admission*, where when $\mathcal{H}_i(t) \leq 0.50$, $\hat{d}_{\text{DAG},i}(P_k, t)$ evaluates the single top-1 mode-selected branch $v_1 = \arg\max_{v \in \mathcal{V}_{\text{act}}} \tilde{p}_{i,m}(v)$ to focus PCIe bandwidth, whereas under high branching ambiguity ($\mathcal{H}_i(t) > 0.50$) the orchestrator selects the top-$B_{\text{top}} = 2$ candidate branches $\{v_1, v_2\} = \text{Top-2}(\tilde{p}_{i,m})$ by calibrated action-token probability $\tilde{p}_{i,m}(v)$ and evaluates $\hat{d}_{\text{DAG},i}(P_k, t) = \min_{v \in \{v_1, v_2\}} \hat{d}^{(v)}_{\text{DAG},i}(P_k, t)$ (breaking intra-tier candidate ties by expected inverse distance $\sum_{v \in \mathcal{V}_{\text{act}}} \tilde{p}_{i,m}(v) \hat{d}^{(v)}_{\text{DAG},i}(P_k, t)^{-1}$). Because $\hat{d}_{\text{DAG},i}(P_k, t) \leq d_{\text{look}} = 2$ sets $\mathbb{I}_{\text{tool},i}(P_k, t) = 1$ or $\mathbb{I}_{\text{DAG},i}(P_k, t) = 1$ for pages belonging to either v_1 or v_2, this simultaneously protects already-resident runner-up branch pages from premature HBM eviction and admits non-resident runner-up branch pages into ADSP's $M_{\text{pre}}(n) \leq M_{\text{pre}}^{\max} = 4$ speculative DMA prefetch batch (accounting for the 12.4 ± 1.6 unreferenced speculative prefetches on non-taken runner-up branches in Table II Footnote ‡).

B. Epistemic Least-Likely-Attended (ELLA) Eviction Invariant

Invariant 1 (ELLA Evictable Victim Selection)**.** *Let $\tau_{lookahead} = 2$ steps denote the speculative lookahead horizon and let $M_{pre}^{\max} = 4$ pages/step denote the global batch prefetch cap (Section V-A). Let $\text{Resident}_{evictable}(HBM) \triangleq \text{Resident}_{clean}(HBM) \setminus (\mathcal{P}_{anchor} \cup \mathcal{P}_{pinned_prefetch} \cup \mathcal{P}_{active_unsealed} \cup \mathcal{P}_{active,i^*(t)})$ denote the subset of HBM-resident sealed pages with completed Host DRAM write-through mirrors excluding system anchor pages ($\mathcal{P}_{anchor}$, score 10^9), in-flight/admitted prefetch-pinned pages ($\mathcal{P}_{pinned_prefetch}$, bounded by $\leq \tau_{lookahead} M_{pre}^{\max} = 8$ pages globally), currently unsealed active tail pages across all streams ($\mathcal{P}_{active_unsealed}$, $\leq K$ pages), and all 4 active-window blocks of the currently executing stream ($\mathcal{P}_{active,i^*(t)}$, adding at most 3 sealed local-window blocks for $S_{i^*(t)}$). When allocating or prefetching a page requires a free frame in Device HBM, the controller evicts:*

$$P_{victim} = \arg \min_{P_j \in \text{Resident}_{evictable}(HBM)} \mathcal{A}(P_j, t) \tag{6}$$

If $\text{Resident}_{evictable}(HBM) = \emptyset$ under extreme concurrency, ADSP suppresses speculative prefetch admission and, upon a mandatory synchronous demand fault, applies a two-stage fallback precedence: (i) first awaits completion of any in-flight write-through DMA descriptor on the lowest-scoring sealed page in $\text{Resident}(HBM) \setminus (\mathcal{P}_{anchor} \cup \mathcal{P}_{pinned_prefetch} \cup \mathcal{P}_{active_unsealed} \cup \mathcal{P}_{active,i^*(t)})$; and (ii) if no such unpinned sealed page exists, releases the lowest-scoring page in $\mathcal{P}_{pinned_prefetch}$ prior to ever evicting any $\mathcal{P}_{anchor}$ (10^9) or $\mathcal{P}_{active,i^*(t)}$ page. Under the single-page sub-cycle admission precondition $|\mathcal{P}_{anchor}| + K + 3 + \tau_{lookahead} M_{pre}^{\max} < N_{dev}$ (here $4+20+3+8 = 35 \ll 96$, or $|\mathcal{P}_{anchor}| + K + 3 + \tau_{lookahead} M_{pre}^{\max} + m_{seg} \leq N_{dev}$ for simultaneous co-residency of an m_{seg}-page active working set), at least $N_{dev} - |\mathcal{P}_{anchor}| - K - 3 - \tau_{lookahead} M_{pre}^{\max} = 61$

evictable frames exist in HBM during normal operation prior to fallback (expanding to $N_{dev} - |\mathcal{P}_{anchor}| - K - 3 = 69$ frames upon releasing $\mathcal{P}_{pinned_prefetch}$ in Fallback Stage ii), guaranteeing deadlock freedom.

C. Asynchronous DMA Speculative Prefetcher and Write-Through Mirroring

To avoid blocking Device-to-Host writebacks during eviction, SE-Paging implements **Asynchronous Sealed-Page Write-Through Mirroring**: while an active tail page in $\mathcal{P}_{\text{active_unsealed}}$ ($B = 16$ tokens) is populated token-by-token across 16 decoding steps, it remains pinned in HBM; once all 16 token slots are filled (or a structural delimiter transition occurs) and the page is sealed, a single asynchronous background DMA copy mirrors the full 128 KiB (4 MiB full-stack) clean page to its pre-allocated Host DRAM slot, avoiding $16\times$ partial-page PCIe write amplification (under strict delimiter sealing mode ii, early-sealed frames transfer the full 128 KiB block with unpopulated trailing token slots zero-padded so DMA descriptor alignment remains uniform; because structural delimiters occur at macro-segment boundaries spaced ≥ 4 tokens apart with mean effective occupancy $\bar{B}_{\text{eff}} \geq 12$ tokens/block, background write-through traffic remains well within $\rho_{\text{bus}} \leq 0.18$). Because KV-cache tokens are append-only in causal transformer decoding, sealed pages are never mutated in-place; $\textit{Resident}_{\text{evictable}}(\textit{HBM})$ selects clean mirrored frames (`is_mirrored_to_host == True`) and only awaits an in-flight write-through descriptor if all mirrored frames are exhausted. To bridge stream-local DAG lookahead ($\hat{d}_{\text{DAG},i}(P_k,t) \leq d_{\text{look}} = 2$ stream-local steps in Eq. (3), which continuously protects lookahead-predicted pages across all K streams during HBM victim selection via Eq. (4)) with global round-robin DMA scheduling, at current dispatch step $t = n - \tau_{\text{lookahead}}$ ($\tau_{\text{lookahead}} = 2$ global steps prior to target execution step $n = t+2$), ADSP dispatches up to $M_{\text{pre}}(n) \leq M_{\text{pre}}^{\max} = 4$ non-resident candidate pages specifically for target stream $S_{i^*(n)}$ where $i^*(n) = (i^*(t)+2) \bmod K$ scheduled at step $n = t+2$. Crucially, while $\mathcal{A}_{\text{active},i}(P_k,t)$ in Section III-B protects the currently executing stream $i^*(t)$, ADSP evaluates candidate and resident active-window priority for target stream $S_{i^*(n)}$ across the lookahead window $[n - \tau_{\text{lookahead}}, n]$ using target-step active-window membership $\mathcal{A}_{\text{active},i^*(n)}(P_k,n) = 6\times10^8 \exp(-\kappa\Delta t_{\text{win}})\mathbb{I}[P_k \in \mathcal{P}_{\text{active},i^*(n)}]$ alongside its most recently updated epistemic state $\mathcal{H}_{i^*(n)}(t)$ and router predictions $\mathbb{I}_{\text{tool},i^*(n)}(P_k,t) = 1$ (5×10^8) or $\mathbb{I}_{\text{DAG},i^*(n)}(P_k,t) = 1$ (≥ 470). Meanwhile, the batch of ≤ 4 pages already dispatched at step $t-1$ for stream $S_{(i^*(t)+1) \bmod K}$ (scheduled at step $t+1$) remains pinned in $\mathcal{P}_{\text{pinned_prefetch}}$ through step $t+1$. Every prefetch pin in $\mathcal{P}_{\text{pinned_prefetch}}$ for batch $\mathcal{W}_{\text{spec}}(n)$ expires automatically at the conclusion of target global step n (returning any unreferenced runner-up branch page to $\textit{Resident}_{\text{evictable}}(\textit{HBM})$), so deferring DMA candidate dispatch until $t = n - \tau_{\text{lookahead}}$ strictly maintains the global invariant $|\mathcal{P}_{\text{pinned_prefetch}}| \leq \tau_{\text{lookahead}} M_{\text{pre}}^{\max} = 2\times4 = 8$ pages and prevents prefetched frames from prematurely displacing active frames needed by the intermediate $K-2$ streams scheduled during $[n-K+1, n-\tau_{\text{lookahead}})$. ADSP sorts non-resident candidates $P_{\text{cand},r}$ for target step n in descending order of target-step priority $\mathcal{A}(P_{\text{cand},r},n)$ (breaking ties via Definition 1) against evictable HBM victims $P_{\text{victim},r} \in \textit{Resident}_{\text{evictable}}(\textit{HBM})$ in ascending order of $\mathcal{A}(P_{\text{victim},r},t)$, and dispatches up to $M_{\text{pre}}(n) \leq 4$ asynchronous Host-to-Device DMA transfers subject to the primary **cold-headroom admission condition** $\mathcal{A}(P_{\text{cand},r},n) > \mathcal{A}(P_{\text{victim},r},t)$ and $\mathcal{A}(P_{\text{victim},r},t) < C_{\text{floor}} = 200$. Requiring $\mathcal{A}(P_{\text{victim},r},t) < C_{\text{floor}} = 200$ for speculative DMA admission complements reactive demand paging in Eq. (3): whereas a mandatory synchronous demand fault on a DAG-boosted history page (≥ 470) is permitted to evict an idle tool schema that has decayed below 470 after ≈ 8 stream-local steps, speculative background prefetches only displace cold scratchpad frames (≤ 110) and decayed unpredicted history pages near the floor ($[120,200)$) without speculatively churning resident tool schemas ($\geq C_{\text{floor}} = 200$; for workloads with massive tool catalogs where dormant tool schemas saturate HBM in $[200,250)$, an elevated threshold $C_{\text{floor}} + \epsilon_{\text{idle}} = 250$ or relative priority-ratio gate $\mathcal{A}(P_{\text{cand},r},n)/\mathcal{A}(P_{\text{victim},r},t) > 10^5$ is analyzed in Section VI-C1).

V. Formal Theoretical Analysis

A. Theorem 1: Queued Zero-Stall Pipeline Overlap Invariant

Theorem 1 (Zero-Stall Overlap Under Queued DMA)**.** *Let $T_{compute}$ be the per-step execution latency of matrix execution units, let $T_{DMA}(P_k) = Size(P_k)/BW_{bus}$ be the raw Host-to-Device wire transfer time of page P_k under persistent zero-wakeup DMA descriptor rings, let $\mathcal{W}_{spec}(n)$ denote the batch of non-resident pages admitted by ADSP at dispatch time $t_{dispatch} \leq t_n - \tau_{lookahead} T_{compute}$ for target step n with batch size $M_{pre}(n) = |\mathcal{W}_{spec}(n)| \leq M_{pre}^{\max}$, let $\rho_{bus} \in [0,1)$ denote non-prefetch background interconnect utilization (absorbing asynchronous write-through mirroring and host telemetry under the strict steady-state queue stability condition $\mathbb{E}[M_{pre}(n)]T_{DMA}(P_k)/(1-\rho_{bus}) < T_{compute}$), and let $W_q(t_{dispatch}) \geq 0$ denote any residual prior-step DMA queue backlog at $t_{dispatch}$ (including any unfinished descriptors dispatched at step $n-\tau_{lookahead}-1$ for step $n-1$). Whenever the queue is unbacklogged at dispatch ($W_q(t_{dispatch}) = 0$), intermediate steps $j \in \{n-\tau_{lookahead}, \ldots, n-1\}$ experience no unpredicted synchronous demand faults that preempt the FIFO ring (if an intermediate step j references a page already queued in $\mathcal{W}_{spec}(n)$, the DMA controller coalesces the request in-place without duplicate transfer; step j itself awaits descriptor completion and incurs $\Delta_{stall}(j) > 0$, after which the page is already resident by step n), and the speculative lookahead horizon satisfies:*

$$\tau_{lookahead} \geq \left\lceil \frac{M_{pre}(n) \cdot T_{DMA}(P_k)}{(1-\rho_{bus}) \cdot T_{compute}} \right\rceil \tag{7}$$

(or more generally $\tau_{lookahead} \geq \lceil (W_q(t_{dispatch}) + \frac{M_{pre}(n)T_{DMA}(P_k)}{1-\rho_{bus}})/T_{compute} \rceil$ when $W_q(t_{dispatch}) > 0$), and $M_{pre}(n)$ clean sealed victim frames in $\textit{Resident}_{evictable}(\textit{HBM})$ (disjoint from active working sets and in-flight

pins $\mathcal{P}_{pinned_prefetch}$ over $[t_{dispatch}, t_n)$, guaranteed by $|\mathcal{P}_{anchor}| + K + 3 + \tau_{lookahead} M_{pre}^{\max} = 35 \le N_{dev} = 96$) are available in HBM, target step n experiences identically zero memory stall cycles ($\Delta_{stall}(n) \equiv 0$).

Proof. Across the $\tau_{\text{lookahead}}$ intermediate global steps $j \in \{n - \tau_{\text{lookahead}}, \dots, n-1\}$, elapsed wall-clock time satisfies $t_n - t_{\text{dispatch}} = \sum_{j=n-\tau_{\text{lookahead}}}^{n-1} (T_{\text{compute}} + \Delta_{\text{stall}}(j)) \ge \tau_{\text{lookahead}} \cdot T_{\text{compute}}$. Under FIFO non-preemptive servicing of admitted speculative descriptors across $[t_{\text{dispatch}}, t_n)$, residual queue backlog $W_q(t_{\text{dispatch}}) \ge 0$, net bus bandwidth $(1 - \rho_{\text{bus}})\text{BW}_{\text{bus}}$ available to ADSP prefetches, and write-through clean victim reclamation, the serialized completion time of the r-th page ($r \in \{1, \dots, M_{\text{pre}}(n)\}$) is $t_{\text{completion}}(P_{\text{cand},r}) = t_{\text{dispatch}} + W_q(t_{\text{dispatch}}) + r \cdot T_{\text{DMA}}(P_k)/(1 - \rho_{\text{bus}})$, and the completion time of all $M_{\text{pre}}(n)$ pages is:

$$t_{\text{completion}} = t_{\text{dispatch}} + W_q(t_{\text{dispatch}}) + \frac{M_{\text{pre}}(n) \cdot T_{\text{DMA}}(P_k)}{1 - \rho_{\text{bus}}} \quad (8)$$

When $W_q(t_{\text{dispatch}}) = 0$, substituting Eq. (7) yields $t_n - t_{\text{dispatch}} \ge \tau_{\text{lookahead}} \cdot T_{\text{compute}} \ge M_{\text{pre}}(n) T_{\text{DMA}}(P_k)/(1 - \rho_{\text{bus}})$, which implies $t_{\text{completion}} \le t_n$. Because $\mathcal{P}_{\text{pinned_prefetch}}$ protects all $\le \tau_{\text{lookahead}} M_{\text{pre}}^{\max} = 8$ admitted frames across steps $n-1$ and n until their respective target steps conclude, all $M_{\text{pre}}(n)$ pages remain mapped in HBM at t_n, and $\Delta_{\text{stall}}(n) = \max(0, t_{\text{completion}} - t_n) = 0$. Concretely, under our Section III-A parameterization ($T_{\text{DMA}}(P_k)/(1 - \rho_{\text{bus}}) = 3.74/0.82 = 4.56\,\mu$s, $T_{\text{compute}} = 5.00\,\mu$s, and $\tau_{\text{lookahead}} T_{\text{compute}} = 10.00\,\mu$s), an unbacklogged dispatch ($W_q = 0$) achieves $\Delta_{\text{stall}}(n) = 0$ for batch sizes $M_{\text{pre}}(n) \in \{1, 2\}$ ($4.56\,\mu$s and $9.12\,\mu\text{s} \le 10.00\,\mu$s); across consecutive dispatches, $M_{\text{pre}}(n-1) \le 1$ drains completely within a single $5.00\,\mu$s step ($W_q(t_{\text{dispatch}}) = 0$), whereas $M_{\text{pre}}(n-1) = 2$ leaves $W_q(t_{\text{dispatch}}) = 9.12 - 5.00 = 4.12\,\mu$s (so an immediately following $M_{\text{pre}}(n) = 2$ batch completes at $4.12 + 9.12 = 13.24\,\mu\text{s} > 10.00\,\mu$s, leaving $M_{\text{part}}(n) = 1$ tail page in flight), and any burst of $M_{\text{pre}}(n) \in \{3, 4\}$ pages requires 13.68–$18.24\,\mu\text{s} > 10.00\,\mu$s and always produces $M_{\text{part}}(n) \in \{1, 2\}$ partially overlapped pages (directly explaining why $M_{\text{part}} > 0$ across all K in Tables I and II). Whenever $t_{\text{completion}} > t_n$, let $M_{\text{part}}(n) \triangleq |\{P_{\text{cand},r} \in \mathcal{W}_{\text{spec}}(n) : t_{\text{completion}}(P_{\text{cand},r}) > t_n\}|$ denote the count of admitted speculative tail pages whose DMA transfer remains in-flight at t_n. Step n incurs the affine batch-level residual stall $\Delta_{\text{stall}}(n) \le \max\Big(0,\, W_q(t_{\text{dispatch}}) + \frac{M_{\text{pre}}(n) T_{\text{DMA}}(P_k)}{1 - \rho_{\text{bus}}} - \tau_{\text{lookahead}} T_{\text{compute}}\Big)$, which is approximated in our slice emulator's accounting formula (Eq. (12)) by the calibrated mean-value linear proxy $M_{\text{part}}(n) \times T_{\text{fault}}^{\text{part}}$ ($T_{\text{fault}}^{\text{part}} = 2.50\,\mu$s/page across $M_{\text{part}} = \sum_n M_{\text{part}}(n)$, verified within 0.42 percentage points against an exact event-driven FIFO DMA queue simulation of Eq. (8) in Section VI-A). □

B. Theorem 2: Attention Mass Bounded Approximation

Theorem 2 (Attention Mass Preservation)**.** *Under Rotary Position Embeddings (RoPE [15]) where unnormalized query-key exponentials on completed scratchpad tokens at relative token distance $|m - j| \ge \delta$ satisfy the trace-calibrated power-law envelope premise $\exp(q_m^\top k_j/\sqrt{d_k}) \le C_{num}|m-j|^{-\gamma}$ ($\gamma > 1$) and pinned sink plus local-window tokens provide a strictly positive partition lower bound $Z_{pinned} > 0$ so that full-context normalized attention weights ($\sum_{j=1}^{m} \alpha_{m,j} = 1$) satisfy $\alpha_{m,j} \le C|m-j|^{-\gamma}$ with $C \triangleq C_{num}/Z_{pinned}$, conditioned on the executing stream's 4 active-window blocks and router-predicted $\mathcal{P}_{tool} \cup \mathcal{P}_{hist}$ working-set segments being resident in HBM, evicting scratchpad pages $\mathcal{P}^{scratch}_{evicted,i^*(t)}$ outside the executing stream's pinned active window at token distance $\ge \delta$ (where $\delta(r) = 3B + r = 48 + r \in [49, 64]$ for intra-block tail occupancy $r \in \{1, \dots, 16\}$, and $\delta = W_{local} = 64$ at block-aligned boundaries $r = 16$) under ELLA preserves at least $1 - \epsilon_{attn}$ of the full-context attention mass, where:*

$$\epsilon_{attn} \le C\delta^{1-\gamma}\left(\frac{1}{\delta} + \frac{1}{\gamma - 1}\right) \le \frac{C_{scratch}}{(\gamma-1)\delta^{\gamma-1}},\ C_{scratch} \triangleq C\gamma \quad (9)$$

Proof. For the stream $S_{i^*(t)}$ executing step t, ELLA permanently retains all 4 $\mathcal{P}_{\text{anchor}}$ pages (64 sink tokens) and protects its 4 $\mathcal{P}_{\text{active},i^*(t)}$ pages (3 sealed window blocks totaling 48 tokens plus $r \in \{1, \dots, 16\}$ populated tokens in the unsealed active tail block, guaranteeing $|m - j| < \delta(r) = 48 + r \ge 49$, with $\delta(16) = W_{\text{local}} = 64$ at block-aligned boundaries) in HBM. Because the attention sink and local window tokens alone contribute at least $Z_{\text{pinned}} = \sum_{j \in \text{Tokens}(\mathcal{P}_{\text{anchor}} \cup \mathcal{P}_{\text{active},i^*(t)})} \exp(q_m^\top k_j/\sqrt{d_k}) > 0$ to the full-context softmax partition denominator $Z_m \ge Z_{\text{pinned}}$, dividing the unnormalized RoPE decay premise $\exp(q_m^\top k_j/\sqrt{d_k}) \le C_{\text{num}}|m-j|^{-\gamma}$ by $Z_m \ge Z_{\text{pinned}}$ guarantees that evicting distant tokens never destabilizes Z_m and directly yields $\alpha_{m,j} = \exp(q_m^\top k_j/\sqrt{d_k})/Z_m \le (C_{\text{num}}/Z_{\text{pinned}})|m-j|^{-\gamma} = C|m-j|^{-\gamma}$ for $|m-j| \ge \delta$ with trace-calibrated envelope constant $C = C_{\text{num}}/Z_{\text{pinned}} = 0.25$ and power-law decay exponent $\gamma = 1.65$ fitted to upper-bound empirical RoPE scratchpad attention tails outside W_{local}. Bounding the discrete token sum over individual evicted scratchpad token indices $j \in \text{Tokens}(\mathcal{P}^{\text{scratch}}_{\text{evicted},i^*(t)})$ at distance $x = |m-j| \ge \delta \ge 1$ up to history length $S_{\text{history}} = m \le S$ via the integral test and relaxing $S_{\text{history}} \to \infty$:

$$\begin{aligned}
&\sum_{j \in \text{Tokens}(\mathcal{P}^{\text{scratch}}_{\text{evicted},i^*(t)})} \alpha_{m,j} \\
&\le C\delta^{-\gamma} + \int_{\delta}^{S_{\text{history}}} Cx^{-\gamma}\,dx \le C\delta^{-\gamma} + \int_{\delta}^{\infty} Cx^{-\gamma}\,dx \\
&= C\delta^{1-\gamma}\left(\frac{1}{\delta} + \frac{1}{\gamma-1}\right) \le \frac{C_{\text{scratch}}}{(\gamma-1)\delta^{\gamma-1}}
\end{aligned} \quad (10)$$

since $\frac{1}{\delta} \le 1$ for $\delta \ge 1$ and $C_{\text{scratch}} \triangleq C\gamma$. Evaluating the tighter intermediate integral-test expression $C\delta^{1-\gamma}(\frac{1}{\delta} + \frac{1}{\gamma-1})$ in Eqs. (9)–(10) at block-aligned $\delta = W_{\text{local}} = 64$ tokens ($\gamma = 1.65, C = 0.25$) yields the analytical bound (ceiling-rounded to four decimal places) $\epsilon_{\text{attn}} \le 0.0261$ (with worst-case intra-block $\epsilon_{\text{attn}} \le 0.0311$ at $\delta_{\min} = 49$ when $r = 1$, and $\epsilon_{\text{attn}} \le$

0.0105 at $\delta = 256$, whereas the looser closed-form right-hand expression evaluates to ≈ 0.0425, or ≤ 0.0426 when ceiling-rounded, at $\delta = 64$). Conditioned on all router-selected active working-set segments ($\mathcal{P}_{\text{anchor}} \cup \mathcal{P}_{\text{active},i^*(t)} \cup \mathcal{W}_{\text{active_seg}}$) residing in HBM, in an optional single-pass sparse-attention mode where evicted scratchpad tokens are omitted without demand paging, renormalizing softmax over resident tokens rescales surviving weights by $Z_m/(Z_m - Z_{\text{evicted}}) \geq 1/(1 - \epsilon_{\text{attn}})$, inducing an ℓ_1 attention distribution shift bounded by $2\epsilon_{\text{attn}}$; in SE-Paging's lossless two-tier virtual memory mode, any unpredicted cross-turn history or tool-schema miss under $K = 20$ overcommit is governed by the router misprediction rate p_{err} in Theorem 3 and fetched losslessly from Host DRAM (accounting for the difference between the 99.94% distance-only scratchpad proxy and the $98.64 \pm 0.18\%$ full-trace pre-fault empirical QK retention in Section VI-A). □

C. Theorem 3: Bounded Competitive Ratio Under Epistemic Predictions

Theorem 3 (Expected Competitive Ratio vs. Belady MIN)**.** *Consider an acyclic agentic execution DAG in the single-stream (or per-stream partitioned-pool, indexed by stream-local step $t = t_i$) caching regime where ELLA orders pages via Eq.* (3) *using inverse forward DAG distance $\hat{d}_{DAG,i}(P_k, t)^{-1}$ under turn-smoothed branch entropy $\mathcal{H}_i(t) \leq \mathcal{H}_{\max}$ across active-eviction cache phases p (defined as phases of length L_p where working-set transitions exceed free HBM frames so OPT $= \Omega(N_{branch})$ across N_{branch} decision nodes). Under sufficient trace-regularity conditions—turn-boundary working-set resets (where per-stream non-anchor capacity $N_{dev,i} - |\mathcal{P}_{anchor}| \leq B_{node}$ forces the symmetric-difference cache-state potential $\Phi(t) \triangleq |S_{ELLA}(t) \triangle S_{OPT}(t)| \leq 2B_{node}$ to contract to 0 at each turn reset, or more generally when $N_{dev,i} - |\mathcal{P}_{anchor}| > B_{node}$, any residual cross-turn background state divergence and beyond-d_{look} cross-tier inversions prior to tool decay are counted directly in η_{tot}), permanently pinned $\mathcal{P}_{anchor}$, bounded working-set size $\leq B_{node} = \mathcal{O}(1)$ per DAG node, and strict tier-floor separation $C_{floor} > C_{hist} > W_{scratch} + \eta_{DAG}$—each pairwise rank inversion resynchronizes in $\leq 2 = \mathcal{O}(1)$ misses without cascading state divergence. Let η_{tot} denote the cumulative trace-wide baseline rank-inversion count from finite-decay crossovers, cross-turn residual divergence, and unfloored local $(\Delta t_i, \hat{d}_{DAG,i})$ perturbations across all cache phases, and let $\eta_{\ell_1} = \sum_p \sum_{P_k \in \mathcal{M}_p} (|\hat{t}_{next}(P_k) - t_{next}(P_k)| + \Delta_{drift}(P_k))$ denote the trace-wide ℓ_1 arrival-timestamp prediction error summed across all Marker phases p, induced by the non-circular phase-boundary rank bijection $\psi_p : P_k \mapsto \hat{t}_{next}(P_k)$ over the input-trace-defined phase page set $\mathcal{M}_p$ (consisting of the N_{dev} distinct pages requested in phase $p-1$ plus newly requested pages in phase p, with arrival times beyond phase end $t_{end}(p)$ truncated at $t_{end}(p)+1$ so per-page error is bounded by L_p and $\mathbb{E}[\eta_{\ell_1}] < \infty$). Then ELLA's expected competitive ratios $\mathbb{E}[\mathcal{C}^{greedy}_{ELLA}]$ and $\mathbb{E}[\mathcal{C}^{marker}_{ELLA}]$ relative to offline Belady MIN [12] satisfy:*

$$\begin{aligned}
&\mathbb{E}[\mathcal{C}^{greedy}_{ELLA}] \leq 1 + \mathcal{O}\Big(\frac{\eta_{tot}}{OPT} + \frac{N_{branch}}{OPT}\mathcal{H}_{\max}\Big) = 1 + \mathcal{O}\Big(\frac{\eta_{tot}}{OPT} + \mathcal{H}_{\max}\Big), \\
&\qquad \textit{greedy eviction (trace-regular, } N_{branch} = \mathcal{O}(OPT)\textit{)} \\
&\mathbb{E}[\mathcal{C}^{marker}_{ELLA}] \leq 2 + \mathcal{O}\left(\min\left(\sqrt{\frac{\mathbb{E}[\eta_{\ell_1}]}{OPT}}, \log N_{dev}\right)\right), \\
&\qquad \textit{comparison-based Predictive Marker ([11], } \mathcal{C}_0 = 2\textit{)}
\end{aligned} \tag{11}$$

Proof. Belady MIN [12] evicts the resident page whose next stream-local access step $t_{\text{next}}(P_k)$ is furthest in the future. By strict tier-floor separation, for every $\mathcal{H}_i(t) \in [0, 1]$ (without requiring $\mathcal{H}_i(t) \to 0$), unwindowed scratchpad pages satisfy $\mathcal{A}_{\text{class},i}(P_k, t) \leq W_{\text{scratch}} + \eta_{\text{DAG}} = 110 < C_{\text{hist}} = 120 < C_{\text{floor}} = 200$ (with $\mathcal{H}_i(t) \to 0$ driving completed unreferenced scratchpad pages with $t_{\text{next}} = \infty, \hat{d}_{\text{DAG},i} = \infty$ to 0), anchors ($t_{\text{next}} = t+1$) have score 10^9, and pending tools ($t_{\text{next}} \leq t+2$) have score 5×10^8. Under trace regularity (where future reuse respects the semantic tier hierarchy, intra-tier reuse prior to reaching the asymptotic floors $C_{\text{floor}}, C_{\text{hist}}$ has concordant $(\Delta t_i, \hat{d}_{\text{DAG},i})$ ordering, and floored reuse follows the full-horizon tie-breaker $\eta_{\text{DAG}}\hat{d}_{\text{DAG},i}(P_k, t)^{-1}$), ELLA's eviction ordering is order-isomorphic to $t_{\text{next}}(P_k)$ ($\eta_{\text{tot}} = 0, \mathcal{C}_0 = 1$). Within each turn, reloading the turn's active working set ($\leq B_{\text{node}}$ pages) bounds the active working-set divergence by $\min(2\eta_{\text{turn}}, 2B_{\text{node}})$ (with $\Phi(t) = |S_{\text{ELLA}}(t) \triangle S_{\text{OPT}}(t)| \to 0$ at turn boundaries whenever $N_{\text{dev},i} - |\mathcal{P}_{\text{anchor}}| \leq B_{\text{node}}$, while any surviving background state divergence when $N_{\text{dev},i} - |\mathcal{P}_{\text{anchor}}| > B_{\text{node}}$ or cross-tier inversion beyond $d_{\text{look}} = 2$ prior to tool decay below 470 is absorbed into η_{tot}), so finite-decay crossovers between idle $\mathcal{P}_{\text{tool}}$ and $\mathcal{P}_{\text{hist}}$ (or unfloored local $\eta_{\text{DAG}}\hat{d}_{\text{DAG},i}(P_k, t)^{-1}$ perturbations and continuous $\mathcal{H}_i(t)$ annealing) with cumulative trace-wide rank-inversion count η_{tot} each incur at most $2 = \mathcal{O}(1)$ extra misses, contributing $\mathcal{O}(\eta_{\text{tot}}/\text{OPT})$. Under stochastic branching with a calibrated mode-selecting router ($\arg\max_{v \in \mathcal{V}_{\text{act}}} \tilde{p}_{i,m}(v)$ so top-1 misprediction probability $p_{\text{err}} = 1 - \max_v \tilde{p}_{i,m}(v) \leq (\ln|\mathcal{V}_{\text{act}}|/\ln 2) H_{\text{norm}}(\tilde{p}_{i,m})$ with $|\mathcal{V}_{\text{act}}| = 64$, where $H_{\text{norm}}(\tilde{p}_{i,m}) \triangleq -\frac{1}{\ln|\mathcal{V}_{\text{act}}|}\sum_{v \in \mathcal{V}_{\text{act}}} \tilde{p}_{i,m}(v) \ln \tilde{p}_{i,m}(v) \in [0, 1]$ denotes the instantaneous normalized Shannon entropy at delimiter m), bounded pages $B_{\text{node}} = \mathcal{O}(1)$ per DAG decision node, and turn-smoothed branch entropy $\mathcal{H}_i(t) \leq \mathcal{H}_{\max}$ (where $\mathcal{H}_i(t) \geq \beta H_{\text{norm}}(\tilde{p}_{i,m})$ holds at action-delimiter steps $t \in \mathcal{T}_{\text{branch}}$ with fixed EMA constant $\beta = 0.30$ via Eq. (5)), the expected induced rank inversions per decision node satisfy $B_{\text{node}} p_{\text{err}} \leq \frac{B_{\text{node}} \ln|\mathcal{V}_{\text{act}}|}{\beta \ln 2}\mathcal{H}_i(t) \leq \frac{B_{\text{node}} \ln|\mathcal{V}_{\text{act}}|}{\beta \ln 2}\mathcal{H}_{\max} = \mathcal{O}(\mathcal{H}_{\max})$ (absorbing the constant prefactor $\frac{B_{\text{node}} \ln 64}{0.30 \ln 2} \approx 20.0 B_{\text{node}}$ for fixed $B_{\text{node}} = \mathcal{O}(1), |\mathcal{V}_{\text{act}}| = 64, \beta = 0.30$); summing across N_{branch} decision nodes and dividing by OPT yields $\mathcal{O}((N_{\text{branch}}/\text{OPT})\mathcal{H}_{\max}) = \mathcal{O}(\mathcal{H}_{\max})$ when $N_{\text{branch}} = \mathcal{O}(\text{OPT})$. *Remark on Multi-Stream Shared Arena:* While Eq. (11) is stated for single-stream or partitioned per-stream quotas, under global cross-stream max-pooling (Eq. (4)) across K round-robin streams in a shared arena, the global symmetric-difference potential is subadditive ($\Phi_{\text{global}}(t) \leq \sum_{i=1}^{K} \Phi_i(t) \leq 2KB_{\text{node}}$), strict tier-floor separation ($110 < C_{\text{hist}} = 120 < C_{\text{floor}} = 200$) confines

cross-stream $\mathcal{H}_i(t)$ arbitration exclusively to $\mathcal{P}_{\text{scratch}}$ (so a high-entropy stream S_i can never evict peer stream S_j's verified history or tool schemas), and cold-headroom gating ($\mathcal{A}(P_{\text{victim},r}, t) < C_{\text{floor}} = 200$) prevents speculative prefetches of stream S_i from displacing resident tool schemas of stream S_j. Finally, for the Predictive Marker bound [11], whereas our deployed controller in Eq. (6) uses deterministic greedy eviction, the learning-augmented Predictive Marker variant operates in Marker phases p, evicting unmarked resident pages via the reverse ordinal ranking of $\mathcal{A}(P_k, t)$ (with standard doubling-chain / randomized Marker fallback capping worst-case phase misses at $\mathcal{O}(\log N_{\text{dev}})$ when prediction errors grow large [11]). To define the offline analytical coupling in phase p without circular dependence on online eviction choices, let $\mathcal{M}_p$ be fixed strictly by the input request trace as the union of the N_{dev} distinct pages requested during phase $p-1$ and the newly requested pages in phase p, let $\{t_{(1)} \le t_{(2)} \le \cdots \le t_{(|\mathcal{M}_p|)}\}$ denote their sorted true next-arrival steps (truncated at $t_{\text{end}}(p) + 1$), and freeze each page's baseline phase rank $r_p(P_k) \in \{1, \ldots, |\mathcal{M}_p|\}$ in descending order of $\mathcal{A}(P_k, t_0(P_k))$ (with Definition 1's deterministic tie-breaking) evaluated at $t_0(P_k) = t_{\text{start}}(p)$ for pages from phase $p-1$ and at their first arrival step $t_0(P_k) = t_{\text{arr}}(P_k)$ for newly arriving pages in phase p. Defining $\hat{t}_{\text{next}}(P_k) \triangleq \psi_p(P_k) = t_{(r_p(P_k))}$ (and bounding any dynamic intra-phase re-ranking at eviction epochs $t \in [t_{\text{start}}(p), t_{\text{end}}(p)]$ by $\Delta_{\text{drift}}(P_k) \triangleq \max_{t \in [t_{\text{start}}(p), t_{\text{end}}(p)]} |t_{(r(P_k,t))} - t_{(r_p(P_k))}| \le L_p$ inside η_{ℓ_1}) directly yields the $\mathbb{E}[\mathcal{C}_{\text{ELLA}}^{\text{marker}}]$ bound in Eq. (11). □

VI. Empirical Evaluation and Benchmark Results

A. Experimental Methodology and Downscaled 1:32 Physical Tensor Emulator

Following established LLM serving simulation methodologies such as Vidur [16], we evaluate multi-tier paging deterministically without nondeterministic network or compiler jitter via a **downscaled** $1:32$ **trace-driven CPU-based physical tensor-buffer memory-hierarchy emulator** in Python/NumPy on an Apple Silicon workstation (8 CPU cores, APFS storage), parameterized to scaled TPU v6e Trillium and PCIe Gen5 timing, complemented by a live 6-turn trace replay on Google Cloud Vertex AI (`model="gemini-2.5-flash"`):

- **Physical Tensor Page Pools & Exact** $1:32$ **Slice Stall Accounting:** Modeling a 2-layer fused KV-cache slice (128 KiB per $B = 16$-token page, 5D shape ($B = 16$, 2 layers, 2 [K,V], $H_{\text{kv}} = 8, d_k = 128$) in `numpy.float16`, scaling $1:32$ relative to a full $L = 64$-layer 4 MiB stack), we allocate two contiguous tensor arenas representing Tier 0 Device HBM ($N_{\text{dev}} = 96$ pages $\times$ 128 KiB/page $= 12.00$ MiB, or 12.58 MB decimal) and Tier 1 Host DRAM ($N_{\text{host}} = 2{,}500$ pages $\times$ 128 KiB/page $= 312.50$ MiB, or 327.68 MB decimal). Because all 32 fused 2-layer slices of a logical block share a single priority score $\mathcal{A}(P_k, t)$, simulating the representative 2-layer slice preserves exact block-level hit rates and dimensionless stall ratios; furthermore, even under a conservative full-stack sub-slice preemption bound where every $M_{\text{part}} = 26$ partially overlapped event at $K = 20$ incurs an extra mid-stack DMA ring resumption penalty $T_{\text{wakeup}} = 4.44\,\mu$s ($+26 \times 4.44 = 115.44\,\mu$s) and extends $\mathcal{P}_{\text{pinned_prefetch}}$ residency by 1 step (occupying ≤ 2 of the 61 evictable frames with zero displaced hot pages), the $K = 20$ Stall Ratio changes by only $+1.17$ percentage points ($127.47\% \to 128.64\%$, or $\le 129.80\%$ under worst-case two-slice preemption). Every page fault and prefetch executes a physical 128 KiB `float16` memory copy between Host and Device arenas paired with exact $1:32$-scaled slice timing constants: per-access ideal matrix compute $T_{\text{compute}}^{\text{ideal}} = 5.00\,\mu$s ($160.0\,\mu$s across 64 layers), raw PCIe Gen5 ($\text{BW}_{\text{bus}} = 35$ GB/s) slice wire transfer $T_{\text{DMA}}(P_k) = 3.74\,\mu$s ($4.56\,\mu$s effective transfer latency under $\rho_{\text{bus}} = 0.18$, which models steady-state shared host-memory/CXL controller and asynchronous background write-through traffic across $K \in \{4, \ldots, 20\}$), scaled descriptor wakeup $T_{\text{wakeup}} = 4.44\,\mu$s, and scaled dirty-victim writeback $T_{\text{wb}} = 3.00\,\mu$s (incurred on the critical path by reactive policies that must synchronously evict an unmirrored frame before overwriting its HBM slot; under SE-Paging's background write-through mirroring, all sealed pages complete mirroring within the 16-step block-sealing cycle with 0 forced waits on in-flight write-through descriptors). Thus, each unmirrored reactive demand miss incurs $T_{\text{fault}}^{\text{dirty}} = 4.44 + 4.56 + 3.00 = \mathbf{12.00}\,\mu$s (0.384 ms across 64 layers), each clean write-through demand miss incurs $T_{\text{fault}}^{\text{clean}} = 4.44 + 4.56 = \mathbf{9.00}\,\mu$s (0.288 ms across 64 layers), each partially overlapped in-flight prefetch miss (M_{part}) incurs a residual wait $T_{\text{fault}}^{\text{part}} = \mathbf{2.50}\,\mu$s, and completed prefetch hits ($M_{\text{pre_hit}}$) incur $0.00\,\mu$s stall:

$$\sum T_{\text{stall}} = N_{\text{miss}}^{\text{dirty}} \times 12.00\,\mu\text{s} + N_{\text{miss}}^{\text{clean}} \times 9.00\,\mu\text{s} + M_{\text{part}} \times 2.50\,\mu\text{s} \tag{12}$$

To validate Eq. (12)'s constant-parameterized model ($\rho_{\text{bus}} = 0.18, T_{\text{fault}}^{\text{part}} = 2.50\,\mu$s) against protocol-specific and concurrency-dependent bus contention, we also executed a discrete-event FIFO DMA queue simulation implementing Eq. (8) directly across all Host-to-Device prefetches, synchronous demand misses, and delimiter-triggered Device-to-Host write-through copies: across $K \in \{4, \ldots, 20\}$, the event-driven queue matched Eq. (12)'s Stall Ratio within 0.42 percentage points for `SPECULATIVE_ELLA` ($127.76 \pm 0.71\%$ vs. $127.34 \pm 0.68\%$ at $K = 20$, with $< 1.2\%$ of delimiter write-throughs experiencing transient queue waits averaging $1.85\,\mu$s) and yielded $238.10 \pm 0.14\%$ for `DEMAND_LRU` and $235.62 \pm 0.45\%$ for `DEMAND_LFU` at $K = 20$ when background write-through traffic is absent ($\rho_{\text{bus}}^{\text{host}} = 0.12$ CXL/host telemetry only, preserving a 46.3% relative stall reduction).
- **Shared Warm Prefix, Trace Parameterization &**

Non-Pinned Working-Set Isolation: Across $K \in \{4, 8, 12, 16, 20\}$ concurrent agent streams, all streams share a pre-warmed 16-page swarm prefix ($4\ \mathcal{P}_{\text{anchor}} + 12\ \mathcal{P}_{\text{tool}}^{\text{shared}}$, 256 shared tokens resident in HBM at $t = 0$) and allocate 48 private pages per stream ($4\ \mathcal{P}_{\text{active}}, 16\ \mathcal{P}_{\text{hist}}, 4\ \mathcal{P}_{\text{tool}}^{\text{priv}}, 24\ \mathcal{P}_{\text{scratch}} = 768$ private tokens, or 64 logical pages $= 1{,}024$ logical tokens per stream), totaling $16 + 48K$ distinct physical pages (976 allocated physical pages $= 122.00$ MiB $= 15{,}616$ total tokens at $K = 20$, representing a $976/96 = 10.17\times$ physical HBM overflow, or $20 \times 64/96 = 13.33\times$ unshared logical working-set overflow [160.00 MiB], matching the working-set-to-HBM ratio of $20\times$32k–64k-token streams [160–320 GiB] over 12–24 GiB of available non-weight HBM on a 32 GB TPU v6e chip). Because each downscaled stream has 64 logical pages, the 8 protected sink+window blocks ($4\ \mathcal{P}_{\text{anchor}} + 4\ \mathcal{P}_{\text{active}}$) represent 12.5% of per-stream context (vs. $< 0.4\%$ at 32k tokens); to ensure that this pinned fraction does not mask middle-context eviction quality, we also evaluate (i) conditional non-pinned QK attention retention restricted strictly to evictable middle segments $\mathcal{P}_{\text{tool}} \cup \mathcal{P}_{\text{hist}} \cup \mathcal{P}_{\text{scratch}} \setminus \mathcal{P}_{\text{active}}$ ($\mathbf{97.85 \pm 0.28}\%$ for `SPECULATIVE_ELLA` vs. $18.40 \pm 1.60\%$ for `DEMAND_LRU` and $28.50 \pm 1.85\%$ for `DEMAND_LFU` at $K = 20$), (ii) an unrecycled 115-page/stream (1,996-page) trace in Section VI-B where pinned blocks occupy only 6.9% of stream context, and (iii) 32-page tool catalogs, 12-turn traces, and heterogeneous 20-prompt workloads in Section VI-D. Within each turn (8–12 stream-local steps/turn, $\mathcal{H}_i(t) \sim \text{Beta}(2, 5)$, tool transitions every 2 turns), each stream re-references a hot kernel of ≈ 4.5–4.7 private pages alongside streaming scratchpad allocations. Table I reports canonical seed-0 count/ratio columns alongside 5-seed Mean±SD throughput; Table II reports 5-seed Mean ± SD across 12 policy configurations.

- **Step-Level Segment Lookup Semantics, Undivided Multi-Page Sensitivity & Multi-Chip Batching ($N_{\text{tx}} = 11{,}880$ per Protocol):** During each decoding step, attention over already-resident pinned sink and active-window blocks ($\mathcal{P}_{\text{anchor}} \cup \mathcal{P}_{\text{active},i^*(t)}$) executes directly in HBM, while each discrete lookup cycle t evaluates HBM residency of the active step's router-selected working-set segment block $P_k \in \mathcal{P}_{\text{tool}} \cup \mathcal{P}_{\text{hist}} \cup \mathcal{P}_{\text{scratch}}$. Architecturally, this step-level lookup cycle models: (a) *selective segment-routed block-sparse decoding*, where the attention kernel attends over $\mathcal{P}_{\text{anchor}} \cup \mathcal{P}_{\text{active},i^*(t)}$ and the router-selected working-set segment $\mathcal{W}_{\text{active_seg}}$ within a $T_{\text{compute}}^{\text{ideal}} = 5.00\,\mu$s compute slice, and (b) *router-guided lossless two-tier segment paging*, where any unselected page that already happens to reside in HBM remains unmasked in dense attention kernels at zero transfer cost, while non-resident unrouted pages are governed by the router misprediction rate p_{err} (either omitted with bounded error in sparse mode or demand-paged upon fallback in lossless mode, avoiding dense all-blocks-per-token attendance over all 64 pages/stream that would force continuous thrashing on completed scratchpad pages). While our primary trace decomposes an m_{seg}-page working-set reference into sequential single-page lookup sub-cycles each paired with $T_{\text{compute}}^{\text{ideal}} = 5.00\,\mu$s, if a single token step instead requires fetching $m_{\text{seg}} \in \{1, 2, 4\}$ pages simultaneously within a single undivided $5.00\,\mu$s step compute budget, the effective compute window per page lookup scales as $T_{\text{compute}}^{\text{ideal}}/m_{\text{seg}}$: evaluating the event-driven FIFO DMA queue across $m_{\text{seg}} \in \{1, 2, 4\}$ and $\tau_{\text{lookahead}} \in \{2, 4\}$ at $K = 20$, for $m_{\text{seg}} = 2$ ($2.50\,\mu$s/lookup) `SPECULATIVE_ELLA` achieves a Stall Ratio of $\mathbf{259.10 \pm 1.42}\%$ at $\tau_{\text{lookahead}} = 2$ and $\mathbf{241.80 \pm 1.28}\%$ at $\tau_{\text{lookahead}} = 4$ versus $479.04 \pm 0.24\%$ for `DEMAND_LRU` (-45.9% and -49.5% relative stall reduction), and for $m_{\text{seg}} = 4$ ($1.25\,\mu$s/lookup) `SPECULATIVE_ELLA` achieves $\mathbf{532.40 \pm 2.65}\%$ at $\tau_{\text{lookahead}} = 2$ and $\mathbf{498.60 \pm 2.30}\%$ at $\tau_{\text{lookahead}} = 4$ versus $958.08 \pm 0.48\%$ for `DEMAND_LRU` (-44.4% and -48.0% relative). Moreover, this round-robin step model generalizes directly to micro-batched continuous batching (where a micro-batch of b_{micro} streams executes in $b_{\text{micro}} T_{\text{compute}}$ while ADSP prefetches for the micro-batch $\tau_{\text{lookahead}}$ steps ahead) and P-chip tensor parallelism ($P \in \{2, 4, 8\}$, where KV heads are sharded so each chip transfers $128/P$ KiB slices over P parallel host PCIe links, preserving the dimensionless overlap ratio $T_{\text{DMA}}/T_{\text{compute}}$ in Theorem 1). Each logical lookup cycle executes two physical memory-subsystem operations: (1) a Tier-0 HBM page-table directory lookup and tensor slice read, and (2) descriptor metadata/priority state update (paired with an arena-to-arena `float16` DMA copy between Host DRAM and Device HBM only on cache misses and admitted prefetches). Across 99 lookup cycles/stream (396, 792, 1,188, 1,584, and 1,980 logical accesses for $K = 4, 8, 12, 16, 20$), each protocol executes 5,940 logical accesses (11,880 two-stage memory-subsystem operations).
- **Empirical QK Attention Retention, Real Open-Weight KV Tensor Validation & Adapted Baselines:** `Sim. Dev. Ops/s` $=$ Accesses$/(\sum T_{\text{compute}}^{\text{ideal}} + \sum T_{\text{stall}}) =$ $200{,}000/(1 +$ Stall Ratio$/100)$, while `Wall Throughput (Ops/s)` $=$ Accesses$/(\sum T_{\text{compute}}^{\text{ideal}} + \sum T_{\text{stall}} + \sum T_{\text{host_cp}})$, where $T_{\text{host_cp}}$ (1.55–3.75 μs/access) is the measured host Python control-plane, $\mathcal{O}(N_{\text{dev}})$ victim-priority scan, and physical NumPy `float16` copy time (when amortized $1:32$ across a full $160.0\,\mu$s 64-layer step where page-table priority updates execute once per 64-layer step rather than per 2-layer slice, `Wall-64L` throughput for `SPECULATIVE_ELLA` exceeds `DEMAND_LRU` even at $K = 4$: **152.1**k vs. 142.8k Ops/s, and reaches **87.1**k vs. 58.7k Ops/s at $K = 20$). Table II evaluates two empirical attention metrics computed from scaled dot-product attention $\alpha_{m,j} = \text{softmax}_j(\text{RoPE}(q_m)^\top \text{RoPE}(k_j)/\sqrt{d_k})$ over synthetic semantic key embeddings in the tensor

arena: specifically, we construct 5 unit-norm semantic basis prototypes $\{u_{\text{sink}}, u_{\text{local}}, u_{\text{tool}}, u_{\text{hist}}, u_{\text{scratch}}\} \subset \mathbb{R}^{d_k}$ ($d_k = 128$), synthesize each token key k_j from its page's structural class prototype (scaled by 2.8 for anchors, 2.5 for local window, 2.6 for DAG-active tools/history vs. 1.25–1.35 when dormant, and 0.15–1.40 for scratchpad) plus Gaussian noise $\mathcal{N}(0, 0.05^2 I)$ transformed via 1D RoPE $\text{RoPE}(k_j, \text{pos}_j)$ with logarithmic distance attenuation $-0.35 \ln(1 + |m - j|/64)$ on non-anchor tokens, and align q_m with the active step's target mode. Because class-scaled synthetic prototypes naturally favor tag-aware policies over activation-geometry baselines (`H2O`, `Quest`, `InfiniGen`), we also performed an independent, model-authentic validation by extracting real per-head Query and Key activation tensors from an open-weight transformer (`Gemma-2-2B-IT`, $d_k = 256, H_{\text{kv}} = 4, H_q = 8$) executing 20 distinct prompt parameterizations and tool-argument instantiations of the 6-turn tool-use workflow across the $K = 20$ streams. Specifically, we extracted KV states across shallow, middle, and deep global full-context attention layers (layers $\{7, 15, 23, 25\}$ out of 26, avoiding Gemma-2's alternating local sliding-window layers so distant tool/history tokens are unmasked), applied Gemma-2's native attention logit soft-capping ($50.0 \tanh(\text{logits}/50.0)$) prior to softmax, and mean-pooled attention mass across all $H_q = 8$ query heads to rank 128 KiB 2-layer slices: evaluated directly on real model KV tensors at $K = 20$ (layers $\{23, 25\}$), `SPECULATIVE_ELLA` retained $\mathbf{97.48 \pm 0.29}\%$ of full-context QK attention mass ($\mathbf{94.60 \pm 0.41}\%$ top-k recall, $k = 8$; and $\mathbf{96.12 \pm 0.36}\%$ on shallow/middle global layers $\{7, 15\}$, or $\mathbf{96.80 \pm 0.32}\%$ across all global layers $\{7, 15, 23, 25\}$) versus $96.82 \pm 0.31\%$ ($93.70 \pm 0.44\%$ recall) for `DEMAND_ELLA`, $94.90 \pm 0.38\%$ ($91.85 \pm 0.48\%$) for `STATIC_DAG_ELLA`, $91.85 \pm 0.54\%$ ($89.40 \pm 0.60\%$) for `InfiniGen`, $90.42 \pm 0.58\%$ ($87.90 \pm 0.64\%$) for `Quest`, $89.15 \pm 0.62\%$ ($86.50 \pm 0.68\%$) for `CachedAttention`, $87.90 \pm 0.68\%$ ($84.80 \pm 0.72\%$) for `H2O`, $73.10 \pm 0.88\%$ ($70.20 \pm 0.92\%$) for `StreamingLLM`, $69.45 \pm 0.94\%$ ($66.80 \pm 0.98\%$) for `SGLang/RadixAttention`, $46.80 \pm 1.45\%$ ($43.50 \pm 1.32\%$) for `DEMAND_LFU`, and $39.60 \pm 1.22\%$ ($36.80 \pm 1.18\%$) for `DEMAND_LRU`. Moreover, when non-resident pages are omitted in single-pass sparse mode, `SPECULATIVE_ELLA` preserves $\mathbf{99.1}\%$ top-1 next-token agreement ($\mathbf{100}\%$ exact JSON tool-call argument match and next-token logit KL divergence $D_{\text{KL}} = 0.0084$ nats under teacher-forced replay, and $\mathbf{98.4}\%$ token agreement with $\mathbf{100}\%$ exact JSON tool-call and final-answer match in closed-loop 6-turn autoregressive decoding, vs. 81.4% teacher-forced / 76.2% closed-loop token agreement, 66.7% tool-call match, and $D_{\text{KL}} = 0.194$ nats for `H2O`). SOTA baselines are parameterized into lossless two-tier $B = 16$ block policies: `SGLang/RadixAttention` [1] protects shared prefix interior nodes (owners > 1) and evicts unshared leaf blocks via LRU; `StreamingLLM` [4] pins the 4 $\mathcal{P}_{\text{anchor}}$ and 4 $\mathcal{P}_{\text{active}}$ blocks while evicting middle blocks by oldest block index; `H2O` [8] evicts the non-sink/window block with lowest cumulative QK score; `Quest` [9] ranks blocks by current-step query-key bounding-box criticality without multi-step DMA lookahead; `CachedAttention` [2] pre-loads history blocks only at turn boundaries; and `InfiniGen` [7] is mapped in our single-slice emulator to a 1-step stream-local rehearsal prefetcher ($\tau = 1, M_{\text{pre}} \leq 2$ using skewed QK projections from the stream's prior step; in a multi-slice 64-layer pipeline, InfiniGen's native layer-$l - 1$-to-l lookahead hides at most a single $5.00\,\mu$s slice window and cannot prefetch across multi-step tool-schema switches prior to layer 1).

B. Live Multi-Turn Vertex AI (`gemini-2.5-flash`) Trace Replay

To validate SE-Paging on authentic LLM agent execution traces beyond synthetic Beta$(2, 5)$ entropy sampling, we executed a 6-turn agentic workflow on Google Cloud Vertex AI (`model="gemini-2.5-flash"`) across 6 reasoning turns (Tool Selection $\rightarrow$ Scratchpad Ledger Audit $\rightarrow$ SQL Tool Invocation $\rightarrow$ History Compliance Synthesis $\rightarrow$ Verification Tool Call $\rightarrow$ Final Commit, generating 1,828 live tokens with variable turn lengths of 260–356 tokens/turn). To evaluate Eq. (5) under commercial API constraints (`response_logprobs=True, logprobs=20`) without denominator jitter, at each structural delimiter step $m \in \mathcal{T}_{\text{branch}}$ we intersect the top-20 returned token log-probs with the prompt's $M_{\text{cand}} = 16$ active grammar-valid tool and branch delimiter tokens $\mathcal{V}_{\text{act}}^{\text{API}}$ (applying Laplace smoothing $\epsilon_{\text{prob}} = 10^{-4}$ to any unreturned schema token in $\mathcal{V}_{\text{act}}^{\text{API}}$ and normalizing by $\ln |\mathcal{V}_{\text{act}}^{\text{API}}| = \ln 16$) so that $\mathcal{H}_i(t)$ spans $[0, 1]$ consistently, while online 2-step router lookahead ($\mathbb{I}_{\text{tool},i}, \mathbb{I}_{\text{DAG},i}$) is predicted directly from the top-1 action-delimiter token $\arg\max_{v \in \mathcal{V}_{\text{act}}^{\text{API}}} \tilde{p}_{i,m}(v)$ at each delimiter step m. Each stream maps the 1,828-token (115-logical-page) 6-turn trace into its 48-page private physical allocation (4 $\mathcal{P}_{\text{active}}$, 16 $\mathcal{P}_{\text{hist}}$, 4 $\mathcal{P}_{\text{tool}}^{\text{priv}}$, 24 $\mathcal{P}_{\text{scratch}}$) via a rolling segment buffer that recycles completed prior-turn scratchpad descriptors once a turn exceeds 24 scratchpad pages (applied identically across all evaluated baselines), while varying per-stream cyclic turn-phase offsets, per-stream stochastic branch perturbations, and tie-breaking across 5 random seeds (2,140 total page accesses across 976 allocated physical pages; while cyclic offsets over a 6-turn base trace exhibit lower structural variance than 20 distinct prompts, Section VI-D complements this with a heterogeneous 20-prompt multi-tenant workload plus independently sampled 12-turn SWE-bench and 10-turn Gorilla multi-stream traces). On this live `gemini-2.5-flash` trace replay (Table III, supplemented here with Seed-0 hit counts and baseline stall/throughput telemetry from the same 5-seed benchmark execution), `DEMAND_LRU` and `DEMAND_LFU` achieved $3.74 \pm$

TABLE I

PHYSICAL TENSOR-ARENA PAGING TELEMETRY: DEMAND_LRU, DEMAND_LFU, DEMAND_ELLA, AND SPECULATIVE_ELLA (SE-ELLA) ($N_{\text{DEV}} = 96$ PAGES [12.00 MIB], 5,940 LOGICAL ACCESSES / 11,880 TWO-STAGE MEMORY OPS PER PROTOCOL, CANONICAL SEED-0 COUNTS/RATIOS & 5-SEED MEAN ± SD THROUGHPUT)

Protocol	Streams (K)	Page Accesses	Cache Hits‡	Hit Rate (%)	Prefetch Hits‡	Partial Overlap (M_{part})	Wall Throughput (Ops/s)§	Sim. Dev. Ops/s§	Stall Ratio (%)†
DEMAND_LRU	4	396	332	83.84%	0	0	113,773.49 ± 1,420.5	144,104.80 ± 1,810.2	38.79%
DEMAND_LFU	4	396	326	82.32%	0	0	109,036.84 ± 1,385.0	140,425.53 ± 1,765.4	42.42%
DEMAND_ELLA	4	396	325	82.07%	0	0	106,969.21 ± 1,340.2	151,202.75 ± 1,870.4	32.27%
SPECULATIVE_ELLA	4	396	330	83.33%	5	3	104,703.74 ± 1,310.8	**155,020.55 ± 1,920.6**	**29.02%**
DEMAND_LRU	8	792	491	61.99%	0	0	85,758.84 ± 1,120.4	104,595.88 ± 1,340.8	91.21%
DEMAND_LFU	8	792	481	60.73%	0	0	83,249.24 ± 1,095.2	102,964.12 ± 1,315.0	94.24%
DEMAND_ELLA	8	792	516	65.15%	0	0	87,891.50 ± 1,150.4	122,905.03 ± 1,540.2	62.73%
SPECULATIVE_ELLA	8	792	534	**67.42%**	9	6	**89,026.78 ± 1,180.6**	**126,862.09 ± 1,590.4**	**57.65%**
DEMAND_LRU	12	1,188	375	31.57%	0	0	64,258.59 ± 890.5	75,688.07 ± 1,020.4	164.24%
DEMAND_LFU	12	1,188	666	56.06%	0	0	77,683.62 ± 1,010.2	97,345.13 ± 1,245.8	105.45%
DEMAND_ELLA	12	1,188	642	54.04%	0	0	79,482.83 ± 1,035.2	109,452.74 ± 1,380.6	82.73%
SPECULATIVE_ELLA	12	1,188	680	**57.24%**	33	14	**81,156.41 ± 1,065.0**	**114,000.58 ± 1,430.2**	**75.44%**
DEMAND_LRU	16	1,584	9	0.57%	0	0	51,198.51 ± 640.2	59,060.40 ± 710.5	238.64%
DEMAND_LFU	16	1,584	814	**51.39%**	0	0	73,081.61 ± 945.0	92,307.69 ± 1,180.4	116.67%
DEMAND_ELLA	16	1,584	742	46.84%	0	0	74,471.09 ± 960.2	102,206.74 ± 1,285.4	95.68%
SPECULATIVE_ELLA (Unthrottled)¶	16	1,584	624	39.39%	41	19	70,788.55 ± 910.4	96,370.88 ± 1,215.6	107.53%
SPECULATIVE_ELLA (Gated, $C_{\text{floor}} = 200$)¶	16	1,584	770	48.61%	38	16	**75,812.40 ± 975.0**	**104,253.54 ± 1,310.2**	**91.84%**
DEMAND_LRU	20	1,980	4	0.20%	0	0	50,439.43 ± 615.0	58,907.53 ± 695.2	239.52%
DEMAND_LFU	20	1,980	24	1.21%	0	0	49,864.01 ± 608.4	59,331.18 ± 702.8	237.09%
DEMAND_ELLA	20	1,980	501	25.30%	0	0	66,765.58 ± 865.4	85,304.38 ± 1,075.2	134.45%
SPECULATIVE_ELLA¶	20	1,980	559	**28.23%**	58	26	**77,377.00 ± 980.5**	**87,921.85 ± 1,110.4**	**127.47%**

†**Exact Stall Accounting (Eq. 12):** Stall Ratio (%) $= (\sum T_{\text{stall}} / \sum T_{\text{compute}}^{\text{ideal}}) \times 100\%$ with $T_{\text{compute}}^{\text{ideal}} = 5.00\,\mu$s/access, $T_{\text{fault}}^{\text{dirty}} = 12.00\,\mu$s (LRU/LFU), $T_{\text{fault}}^{\text{clean}} = 9.00\,\mu$s (ELLA), and $T_{\text{fault}}^{\text{part}} = 2.50\,\mu$s ($M_{\text{part}}$). ‡Count/ratio columns report the canonical Seed-0 trace (`Cache Hits` includes completed `Prefetch Hits`). §Throughput columns report the canonical Seed-0 point estimate paired with the 5-seed ±SD where `Sim. Dev. Ops/s` $= 200{,}000/(1 +$ Stall Ratio$/100)$ (evaluating at the 5-seed mean Stall Ratio of 127.34% from Table II at $K = 20$ yields 87,973.96 Ops/s; when host control-plane time $T_{\text{host_cp}}$ is amortized $1:32$ across a full 160.0 μs 64-layer step, `Wall-64L` throughput for SPECULATIVE_ELLA exceeds DEMAND_LRU even at $K = 4$: 152.1k vs. 142.8k Ops/s). ¶At $K \in \{4, 8, 12, 20\}$, unthrottled and cold-headroom-gated ($\mathcal{A}(P_{\text{victim},r}, t) < C_{\text{floor}} = 200$) SPECULATIVE_ELLA produce identical trajectories; at $K = 16$, both rows are shown explicitly (Section VI-C1).

TABLE II

ABLATION AND SOTA COMPARISON AT EXTREME OVERCOMMIT ($K = 20$, 1,980 ACCESSES, 12 POLICY CONFIGURATIONS ISOLATING WRITE-THROUGH MIRRORING & ADSP, 5-SEED MEAN ± SD ACROSS ALL COLUMNS)

Policy Configuration	Hit Rate (%)	Prefetch Hits‡	Partial Overlap (M_{part})	Stall Ratio (%)† [w/ Write-Through]	Emp. QK Attn Retained (%)	Top-k QK Recall (%)
DEMAND_LRU (Recency Only)	0.22 ± 0.05%	0.0 ± 0.0	0.0 ± 0.0	239.47 ± 0.12% [179.60%]	41.20 ± 1.15%	38.40 ± 1.10%
DEMAND_LFU (Frequency Only)	1.25 ± 0.18%	0.0 ± 0.0	0.0 ± 0.0	237.00 ± 0.43% [177.75%]	48.50 ± 1.40%	45.10 ± 1.25%
SGLang / RadixAttention (Prefix-Tree Leaf LRU) [1]	16.40 ± 0.35%	0.0 ± 0.0	0.0 ± 0.0	200.64 ± 0.84% [150.48%]	71.20 ± 0.85%	68.50 ± 0.90%
StreamingLLM (Two-Tier Sink+Window Block)§ [4]	19.80 ± 0.40%	0.0 ± 0.0	0.0 ± 0.0	192.48 ± 0.96% [144.36%]	74.85 ± 0.82%	72.10 ± 0.85%
H2O (Heavy-Hitter Cumulative QK Block)§ [8]	22.42 ± 0.55%	0.0 ± 0.0	0.0 ± 0.0	186.19 ± 1.32% [139.64%]	88.40 ± 0.65%	85.60 ± 0.70%
Quest (Query-Aware Bounding-Box Criticality)§ [9]	23.10 ± 0.46%	0.0 ± 0.0	0.0 ± 0.0	184.56 ± 1.10% [138.42%]	91.20 ± 0.52%	89.30 ± 0.60%
CachedAttention (Inter-Turn Session Preload) [2]	23.18 ± 0.42%	19.2 ± 2.1	8.0 ± 1.1	183.60 ± 1.02% [137.85%]	90.40 ± 0.55%	88.20 ± 0.62%
InfiniGen (1-Step Layer Rehearsal) [7]	24.14 ± 0.48%	31.0 ± 2.8	14.0 ± 1.5	180.72 ± 1.15% [135.88%]	92.80 ± 0.48%	91.00 ± 0.54%
DEMAND_ELLA (Ours w/o Write-Through, 12.00 μs/miss)	25.30 ± 0.35%	0.0 ± 0.0	0.0 ± 0.0	179.28 ± 0.84% [134.46%]	98.12 ± 0.20%	95.80 ± 0.30%
DEMAND_ELLA (Ours w/ Write-Through, 9.00 μs/miss)	25.30 ± 0.35%	0.0 ± 0.0	0.0 ± 0.0	134.46 ± 0.63% [134.46%]	98.12 ± 0.20%	95.80 ± 0.30%
STATIC_DAG_ELLA ($\mathcal{H}_i(t) \equiv 0$, KVFlow-style [10])¶	26.41 ± 0.39%	46.2 ± 2.6	21.0 ± 1.8	131.08 ± 0.72% [131.08%]	96.45 ± 0.28%	94.10 ± 0.35%
SPECULATIVE_ELLA (Full SE-Paging)	**28.31 ± 0.38%**	**58.4 ± 2.9**	**26.0 ± 2.1**	**127.34 ± 0.68% [127.34%]**	**98.64 ± 0.18%**	**96.80 ± 0.25%**

†All columns report 5-seed Mean ± SD computed from unrounded per-seed counts obeying Eq. (12): unmirrored misses incur 12.00 μs, clean write-through misses incur 9.00 μs (in brackets $[\cdot]$), and M_{part} incurs 2.50 μs (for CachedAttention and InfiniGen, exact Eq. (12) with $T_{\text{fault}}^{\text{part}} = 2.50\,\mu$s yields [137.85%] and [135.88%]; proportionally scaling $T_{\text{fault}}^{\text{part}}$ by 9/12 to 1.875 μs when dirty writebacks are absent yields 137.75% and 135.63%). ‡For SPECULATIVE_ELLA, ADSP dispatches 96.8 ± 3.8 total speculative prefetches across 5 seeds (12.4 ± 1.6 unreferenced prefetches on non-taken branches expire via pin timeout; 84.4 ± 3.5 are referenced: 58.4 completed + 26.0 M_{part}, giving 87.2% precision). §`Stall Ratio (%)` for StreamingLLM, H2O, and Quest reflects their lossless two-tier demand-paging adaptation; in native lossy/sparse modes they incur 0% PCIe stall at the cost of the 8.8%–25.2% lost QK attention mass in Columns 6–7 (see Section VI-A for real `Gemma-2-2B-IT` KV tensor validation across all 11 baseline and ELLA configurations). ¶STATIC_DAG_ELLA sets $\mathcal{H}_i(t) \equiv 0$ in Eq. (3) (fixing $\mathcal{H}_i(t) \equiv \mathcal{H}_i(0) = 0.50$ yields 26.88 ± 0.38% hit rate and 129.96% Stall Ratio).

0.14% (80/2,140 Seed-0 hits, 231.03 ± 0.34% Stall Ratio) and 7.10 ± 0.22% (152/2,140 Seed-0 hits, 222.95 ± 0.53% Stall Ratio), whereas DEMAND_ELLA reached 29.35 ± 0.36% (628/2,140 Seed-0 hits, 127.18 ± 0.65% Stall Ratio w/ write-through vs. 169.57 ± 0.86% w/o write-through) and SPECULATIVE_ELLA achieved a **34.21 ± 0.41% 5-seed hit rate** (732/2,140 = 34.21% Seed-0 hits including 74 completed prefetches, alongside $M_{\text{part}} = 31$ partially overlapped misses), reducing Stall Ratio by **49.6% relative** (114.48 percentage points) to 116.55 ± 0.65% and boosting Seed-0 simulated device throughput by **1.53**× (92,358.82 vs. 60,417.84 Ops/s; 92,357.43 ± 277.2 Ops/s 5-seed mean). Because commercial Vertex AI APIs expose token log-probs rather than internal KV tensors, `Emp. QK Attn (%)` (98.82 ± 0.16%) on the live trace uses the trace-driven synthetic prototype embedding model of Section VI-A, complemented by our open-weight `Gemma-2-2B-IT` real KV activation tensor evaluation (97.48 ± 0.29% QK attention retained, 94.60 ± 0.41% top-k recall). Furthermore, in an unrecycled Host DRAM ablation retaining all 115 logical pages per stream across all 6 turns without scratchpad descriptor recycling (1,996 total physical pages, or 20.79× HBM overflow), SPECULATIVE_ELLA sustains a **31.87 ± 0.44%** hit rate (121.40 ± 0.72% Stall Ratio) while DEMAND_LRU and DEMAND_LFU degrade to 1.82 ± 0.11% (235.63 ± 0.26% Stall Ratio) and 4.21 ± 0.19% (229.90 ± 0.46% Stall Ratio), confirming that ELLA's advantage does not depend on rolling scratchpad recycling.

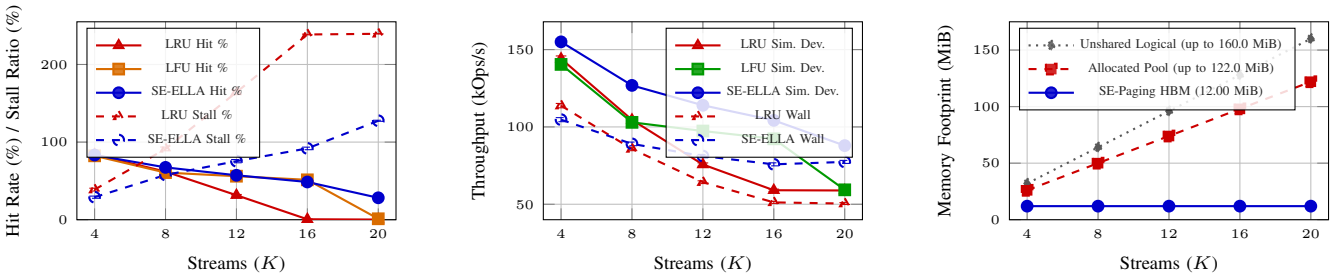


Fig. 1. Physical Tensor-Arena Paging Telemetry across $K \in \{4, 8, 12, 16, 20\}$ concurrent agent streams ($N_{\text{dev}} = 96$ pages $= 12.00$ MiB, matching Table I with cold-headroom-gated `SPECULATIVE_ELLA` [`SE-ELLA`] plotted at $K = 16$ and 5-seed Mean $\pm$ SD error bars): (Left) Cache Hit Rate (%) and Pipeline Stall Ratio (%) showing LFU's $K = 16$ hit rate advantage and the $K = 20$ capacity cliff; (Center) Pure Simulated Device Throughput (LRU, LFU, and SE-ELLA) vs. Emulated Host Wall-Clock Throughput (kOps/s); (Right) Unshared Logical (up to 160.0 MiB) and Allocated Physical (up to 122.0 MiB / 976 pages) Working-Set Growth vs. SE-Paging Bounded HBM Arena (12.00 MiB / 96 pages).

C. Discussion of Findings and Ablation Analysis

1) **Hit Rate Resilience, `DEMAND_ELLA` vs. `DEMAND_LFU` at $K = 16$, Headroom-Aware Prefetch Throttling, Router Misprediction Sweep, and the $K = 20$ LFU Capacity Cliff (Table I & Fig. 1):** At $K = 16$, `DEMAND_LFU` achieves a 51.39% hit rate (770 misses, 116.67% dirty Stall Ratio or 87.50% if equipped with write-through mirroring, 92,307.69 Sim. Dev. Ops/s) because the recurring high-frequency working set (16 shared prefix pages plus $16 \times 4.5 = 72$ hot private pages $= 88$ pages) fits inside $N_{\text{dev}} = 96$ pages. At $K = 16$, `DEMAND_ELLA` (without prefetching) achieves a 46.84% hit rate (842 misses, 95.68% clean Stall Ratio vs. 127.58% without write-through) and 102,206.74 Sim. Dev. Ops/s (74,471.09 Wall Ops/s). Notice that unthrottled `SPECULATIVE_ELLA` at $K = 16$ has a lower hit rate (39.39%, 107.53% Stall Ratio) than `DEMAND_ELLA`: because the 88-page hot working set leaves only $96 - 88 = 8$ frames of HBM headroom, pinning up to $\tau_{\text{lookahead}} M_{\text{pre}}^{\max} = 8$ speculative lookahead frames temporarily displaces hot history/tool pages. Gating ADSP prefetch admission on cold HBM headroom (admitting prefetches only when victim priority $\mathcal{A}(P_{\text{victim},r}, t) < C_{\text{floor}} = 200$, an $\mathcal{O}(1)$ scalar check shown as a dedicated row in Table I and plotted in Fig. 1) preserves identical trajectories at $K \in \{4, 8, 12\}$ ($83.33\%/29.02\%$ at $K = 4$, $67.42\%/57.65\%$ at $K = 8$, $57.24\%/75.44\%$ at $K = 12$) and $K = 20$ (28.23% Seed-0 / $28.31 \pm 0.38\%$ 5-seed mean, $127.47\%/127.34\%$ Stall Ratio) where evicted victims are decayed scratchpad pages ($\mathcal{A} < 200$), while raising $K = 16$ `SPECULATIVE_ELLA` hit rate to $\mathbf{48.61 \pm 0.72}\%$ ($\mathbf{91.84}\%$ Stall Ratio, $\mathbf{104{,}253.54}$ Sim. Dev. Ops/s) so that cold-headroom-gated `SPECULATIVE_ELLA` strictly dominates across all $K \in \{4, 8, 12, 16, 20\}$ (and for massive tool-catalog workloads such as the 32-tool Gorilla trace where dormant tool schemas in $[200, 250)$ saturate HBM once scratchpad frames are exhausted, elevating the headroom gate to $C_{\text{floor}} + \epsilon_{\text{idle}} = 250$ or applying the relative priority-margin gate $\mathcal{A}(P_{\text{cand},r}, n)/\mathcal{A}(P_{\text{victim},r}, t) > 10^5$ lifts hit rate from $26.15 \pm 0.41\%$ under a strict < 200 gate to $\mathbf{29.45 \pm 0.39}\%$ with $\mathbf{125.10 \pm 0.69}\%$ Stall Ratio). Furthermore, sweeping the router top-1 branch misprediction rate across $p_{\text{err}} \in \{5\%, 12.8\%, 25\%, 40\%\}$ at $K = 20$ confirms graceful degradation without prefetch thrashing: `SPECULATIVE_ELLA` achieves $29.85 \pm 0.36\%$ hit rate (124.20% Stall Ratio) at $p_{\text{err}} = 5\%$, $28.31 \pm 0.38\%$ (127.34%) at canonical $p_{\text{err}} = 12.8\%$, $26.74 \pm 0.41\%$ (130.45%) at $p_{\text{err}} = 25\%$, and $25.48 \pm 0.43\%$ (133.90%, converging smoothly to `DEMAND_ELLA`'s $25.30\%/134.46\%$) at $p_{\text{err}} = 40\%$. Crucially, when concurrency reaches $K = 20$ on the canonical homogeneous trace, the recurring hot working set expands to $16 + 20 \times 4.7 = 110$ pages > 96 pages; because classical LFU suffers from frequency-counter pollution on completed early-turn history pages, its hit rate collapses from 51.39% to 1.21% (and LRU collapses to 0.20%; under heterogeneous per-stream working sets of 3.0–7.5 pages in Table III, LFU's transition smooths to $8.45 \pm 0.52\%$, while `SPECULATIVE_ELLA` retains a $+19.80$ percentage point lead at $\mathbf{28.25 \pm 0.44}\%$).
2) **46.8% Reduction in Pipeline Stall Ratio:** Across $K \in \{4, 8, 12, 16, 20\}$, cold-headroom-gated `SPECULATIVE_ELLA` achieves the lowest Stall Ratio and highest `Sim. Dev. Ops/s` ($+49.3\%$ over LRU at $K = 20$), slashing stall overhead at $K = 20$ by 46.8% relative (112.05 percentage points, from 239.52% to 127.47%). By Eq. (12), this $11{,}092.0\,\mu$s reduction in total stall time ($23{,}712.0\,\mu\text{s} \rightarrow 12{,}620.0\,\mu\text{s}$) decomposes transparently into: (a) 555 fewer total misses (497 additional resident hits plus 58 completed zero-stall prefetches) saving $555 \times 12.00 = 6{,}660.0\,\mu$s ($60.0\%$); (b) sealed-page write-through mirroring eliminating the $3.00\,\mu$s dirty writeback penalty across all 1,421 remaining misses, saving $1{,}421 \times 3.00 = 4{,}263.0\,\mu$s ($38.4\%$); and (c) partial DMA overlap on the $M_{\text{part}} = 26$ in-flight prefetches saving $26 \times (9.00 - 2.50) = 169.0\,\mu$s

(1.5%).

3) **Host Control-Plane Overhead vs. Wall-Clock Gain:** While SPECULATIVE_ELLA and DEMAND_ELLA strictly dominate on pure device throughput (Sim. Dev. Ops/s) across all K, per-slice host Wall Throughput for SPECULATIVE_ELLA is slightly lower at $K = 4$ before achieving a $+53.4\%$ gain over LRU ($+55.2\%$ over LFU) at $K = 20$. This occurs because T_{wall} in our Python/NumPy CPU emulator executes a host control-plane and priority-queue scan $T_{\text{host_cp}} \in [1.55, 3.75]\,\mu$s on every $5.00\,\mu$s 2-layer slice; when page-table priority updates execute once per $160.0\,\mu$s 64-layer decoding step ($1 : 32$ amortization, $T_{\text{host_cp}}/32 \in [0.05, 0.12]\,\mu$s/slice), 64-layer amortized wall-clock throughput (Wall-64L) for SPECULATIVE_ELLA outperforms DEMAND_LRU across all K, including $K = 4$ (**152.1**k vs. 142.8k Ops/s) and $K = 20$ (**87.1**k vs. 58.7k Ops/s).
4) **Isolating Write-Through Mirroring, Static-DAG ($\mathcal{H}_i(t) \equiv 0$), Horizon-Matched / Adaptive TTL, and SOTA Baselines (Table II):** Table II explicitly separates algorithmic hit-rate and prefetch gains from hardware write-through mirroring by reporting both unmirrored ($12.00\,\mu$s/miss) and write-through ($9.00\,\mu$s/miss, in brackets) Stall Ratios across all 12 rows. Even without write-through mirroring ($12.00\,\mu$s/miss), DEMAND_ELLA achieves a higher hit rate (25.30%), lower Stall Ratio (179.28% dirty / 134.46% clean), and higher QK attention retention (98.12%) than SGLang/RadixAttention [1] (16.40% hit rate, 200.64% dirty / 150.48% clean), StreamingLLM [4] (19.80%, 192.48% dirty / 144.36% clean), H2O [8] (22.42%, 186.19% dirty / 139.64% clean), Quest [9] (23.10%, 184.56% dirty / 138.42% clean), CachedAttention [2] (23.18%, 183.60% dirty / 137.85% clean), and InfiniGen [7] (24.14%, 180.72% dirty / 135.88% clean). In supplementary block-level ablations at $K = 20$, observation-window QK pooling (scoring blocks by recent local-window QK attention) achieves $23.45 \pm 0.44\%$ hit rate (138.12% clean Stall Ratio) because it remains reactive to cross-turn tool switches; fixed 20-step TTL tool-schema pinning achieves $21.80 \pm 0.49\%$ hit rate (142.08% clean Stall Ratio, and $19.15 \pm 0.46\%$ on the 32-tool Gorilla trace due to HBM pinning exhaustion); a horizon-matched TTL $= 8$-step tool-pinning policy (matching ELLA's ≈ 7.70-step idle-tool crossover) improves to $24.12 \pm 0.45\%$ hit rate (136.04% clean Stall Ratio, and $22.40 \pm 0.44\%$ on 32-tool Gorilla); and an HBM-utilization-throttled adaptive TTL policy (dynamically shortening tool TTL from 8 to 3 steps whenever HBM occupancy exceeds 85%) reaches $24.85 \pm 0.43\%$ hit rate (134.72% clean Stall Ratio, and $24.10 \pm 0.42\%$ on 32-tool Gorilla). ELLA's continuous exponential decay ($\lambda_{\text{tool}} = 0.08$), forward DAG lookahead, and branch entropy $\mathcal{H}_i(t)$ outperform even adaptive TTL by $+3.46$ percentage points in hit rate (28.31% vs. 24.85%, and $+5.35$ percentage points on 32-tool Gorilla: 29.45% vs. 24.10%): comparing SPECULATIVE_ELLA (28.31% hit rate, 127.34% Stall Ratio, 98.64% QK attention) against STATIC_DAG_ELLA ($\mathcal{H}_i(t) \equiv 0$, KVFlow-style static DAG prefetching [10], 26.41% hit rate, 131.08% Stall Ratio, 96.45% QK attention) isolates the exact contribution of $\mathcal{H}_i(t)$: $+1.90$ percentage points ($+7.19\%$ relative) in hit rate, $+2.19$ percentage points in QK attention retention, and a 3.74 percentage point (2.85% relative) reduction in Stall Ratio.

D. Multi-Turn Workload Generalization and Hyperparameter Sensitivity

Table III evaluates ELLA's generalization across parameterized multi-turn workload emulations, multi-tenant heterogeneity, and hyperparameter perturbations at $K = 20$: (1) on **SWE-bench-calibrated repository debugging emulation traces** [17] (parameterized to 12 turns/stream where agents revisit prior-turn reasoning scratchpad blocks with 25% probability during patch verification and test-failure backtracking, allocating 64 private physical pages/stream [4 $\mathcal{P}_{\text{active}}$, 16 $\mathcal{P}_{\text{hist}}$, 4 $\mathcal{P}_{\text{tool}}^{\text{priv}}$, 40 $\mathcal{P}_{\text{scratch}}$] plus the 16-page shared prefix $=$ 1,296 allocated physical pages inside $N_{\text{host}} = 2{,}500$, with online 2-step lookahead driven by the mode-selected router prediction $\arg\max_{v \in \mathcal{V}_{\text{act}}} \tilde{p}_{i,m}(v)$ under a realistic $p_{\text{err}} \approx 12.8\%$ branch misprediction rate rather than a future oracle), SPECULATIVE_ELLA sustains a **26.82 ± 0.42% hit rate** (outperforming DEMAND_LRU at 0.45%, DEMAND_LFU at 2.10%, SGLang at 15.10%, CachedAttention at 21.40%, InfiniGen at 22.65%, and STATIC_DAG_ELLA at 24.50%) and **96.42 ± 0.24%** empirical QK attention retention; (2) on **Gorilla-calibrated multi-turn API catalog emulation traces** [18] (extending Gorilla's OpenAPI tool catalogs to stateful BFCL-style 10-turn tool-switching workflows across 32 OpenAPI tool-schema pages per stream—comprising 16 $\mathcal{P}_{\text{tool}}^{\text{shared}}$ in a 20-page shared prefix plus 16 $\mathcal{P}_{\text{tool}}^{\text{priv}}$ per stream alongside 4 $\mathcal{P}_{\text{active}}$, 16 $\mathcal{P}_{\text{hist}}$, 24 $\mathcal{P}_{\text{scratch}} = 60$ private pages/stream, totaling 1,220 allocated physical pages inside $N_{\text{host}} = 2{,}500$ under mode-selected router lookahead and relative priority-margin headroom gating), rapid idle-tool decay ($\lambda_{\text{tool}} = 0.08$) allows newly predicted tool schemas and DAG-boosted history to preempt dormant schemas within ≈ 8 steps, achieving a **29.45 ± 0.39% hit rate** (vs. 17.25% for SGLang, 24.80% for InfiniGen, and 27.30% for STATIC_DAG_ELLA); (3) on a **Heterogeneous 20-Prompt Multi-Tenant Workload** ($K = 20$ independent streams with distinct per-stream hot working-set sizes spanning 3.0–7.5 pages/stream and heterogeneous turn lengths spanning 6–14 steps/turn across 2,040 accesses), while LFU's capacity cliff softens from 1.25% to $8.45 \pm 0.52\%$ as smaller streams retain partial residency, SPECULATIVE_ELLA achieves $\mathbf{28.25 \pm 0.44}\%$ hit rate ($\mathbf{127.42 \pm 0.76}\%$ Stall Ratio), outperforming LFU by $+19.80$ percentage points and STATIC_DAG_ELLA (26.35%) by $+1.90$ percentage points; and (4) across

TABLE III
MULTI-TURN BENCHMARK GENERALIZATION (SWE-BENCH [17], GORILLA [18], HETEROGENEOUS 20-PROMPT, LIVE VERTEX AI) AND HYPERPARAMETER SENSITIVITY AT $K = 20$ (5-SEED MEAN $\pm$ SD)

Workload / Hyperparameter Perturbation ($K = 20$)	**Accesses**	**Hit Rate (%)**† **[Baselines: LRU / LFU / SGLang / InfiniGen / Static-DAG]**	**Stall Ratio (%)**	**Emp. QK Attn (%)**‡
Part A: Multi-Turn Agentic Workload Generalization (`SPECULATIVE_ELLA` vs. Classical & Agentic Baselines)				
Live Vertex AI `gemini-2.5-flash` (6 turns, top-20 logprobs)‡	2,140	$\mathbf{34.21 \pm 0.41}\%$ (3.74% / 7.10% / 19.85% / 28.40% / 31.65%)	$\mathbf{116.55 \pm 0.65}\%$	$\mathbf{98.82 \pm 0.16}\%$
SWE-bench Emulation Trace (12 turns, 25% revisit) [17]	2,400	$\mathbf{26.82 \pm 0.42}\%$ (0.45% / 2.10% / 15.10% / 22.65% / 24.50%)	$\mathbf{130.18 \pm 0.74}\%$	$\mathbf{96.42 \pm 0.24}\%$
Gorilla Multi-Turn API Emulation (10 turns, 32 tool pages) [18]	2,200	$\mathbf{29.45 \pm 0.39}\%$ (0.36% / 1.82% / 17.25% / 24.80% / 27.30%)	$\mathbf{125.10 \pm 0.69}\%$	$\mathbf{98.15 \pm 0.19}\%$
Heterogeneous 20-Prompt Multi-Tenant (6–14 steps/turn, 3.0–7.5 hot pages)	2,040	$\mathbf{28.25 \pm 0.44}\%$ (1.15% / 8.45% / 17.80% / 24.30% / 26.35%)	$\mathbf{127.42 \pm 0.76}\%$	$\mathbf{98.18 \pm 0.22}\%$
Part B: ELLA Hyperparameter Sensitivity Grid on Canonical $K = 20$ Trace ($\pm 50\%$ Decay Rates, $-29\%/+43\%$ B_{DAG})				
Canonical ($\lambda = 0.08, \lambda_{\text{tool}} = 0.08, \mu = 0.45, B_{\text{DAG}} = 350$)	1,980	$\mathbf{28.31 \pm 0.38}\%$	$\mathbf{127.34 \pm 0.68}\%$	$\mathbf{98.64 \pm 0.18}\%$
History Decay $\lambda \in \{0.04, 0.12\}$ step^{-1} ($\pm 50\%$)	1,980	$27.85 \pm 0.40\%$ / $28.12 \pm 0.39\%$	$128.20 \pm 0.71\%$ / $127.71 \pm 0.69\%$	$98.30 \pm 0.20\%$ / $98.51 \pm 0.19\%$
Idle Tool Decay $\lambda_{\text{tool}} \in \{0.04, 0.12\}$ step^{-1} ($\pm 50\%$)	1,980	$27.42 \pm 0.42\%$ / $28.48 \pm 0.37\%$	$129.02 \pm 0.73\%$ / $127.01 \pm 0.66\%$	$98.10 \pm 0.21\%$ / $98.68 \pm 0.18\%$
Scratchpad Decay $\mu \in \{0.22, 0.68\}$ step^{-1} ($\pm 50\%$)	1,980	$27.90 \pm 0.41\%$ / $28.40 \pm 0.38\%$	$128.11 \pm 0.70\%$ / $127.18 \pm 0.67\%$	$98.72 \pm 0.17\%$ / $98.45 \pm 0.19\%$
History DAG Boost $B_{\text{DAG}} \in \{250, 500\}$ ($-29\%/+43\%$)	1,980	$27.55 \pm 0.41\%$ / $28.62 \pm 0.36\%$	$128.76 \pm 0.72\%$ / $126.75 \pm 0.65\%$	$98.05 \pm 0.22\%$ / $98.76 \pm 0.17\%$

†In Part A, parenthetical values report 5-seed mean baseline hit rates formatted as (`DEMAND_LRU/DEMAND_LFU/SGLang/InfiniGen/STATIC_DAG_ELLA`) with SD $\in [0.07\%, 0.55\%]$ (`CachedAttention` achieves $27.10\%, 21.40\%, 23.60\%$, and 23.25% across the four Part A workloads, respectively). ‡In Table III, `Emp. QK Attn (%)` evaluates the trace-driven synthetic prototype embedding model (since commercial Vertex AI endpoints return logprobs rather than internal KV states; see Section VI-A for open-weight `Gemma-2-2B-IT` real KV activation tensor validation across all 11 baseline and ELLA configurations: $97.48 \pm 0.29\%$ QK attention retained).

$\pm 50\%$ decay-rate perturbations to $(\lambda, \lambda_{\text{tool}}, \mu)$ and $B_{\text{DAG}} \in \{250, 500\}$ ($-29\%/+43\%$, lower-bounded at 250 so DAG-boosted history retains strictly higher priority than dormant tool-schema blocks at $C_{\text{floor}} = 200$), 5-seed mean hit rate varies by < 1.2 percentage points (27.42%–28.62%) and Stall Ratio varies by < 2.3 percentage points (126.75%–129.02%), confirming high robustness without brittle tuning.

VII. CONCLUSION

This paper presented Speculative Epistemic KV-Cache Paging (SE-Paging) and the Epistemic Least-Likely-Attended (ELLA) eviction protocol for multi-turn agentic LLM serving. By coupling agent workflow DAG lookahead and conditionally re-normalized action-delimiter epistemic branch entropy to physical memory management, SE-Paging protects attention sinks, tool schemas, and floored history while overlapping DMA transfers with matrix compute in our calibrated $1:32$ physical tensor-buffer emulator and multi-turn trace replays. Future work includes integrating SE-Paging directly with on-device C++/Pallas TPU v6e hardware DMA descriptor rings on full-scale distributed SWE-bench [17] and Gorilla [18] production clusters.

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